Onderwijsgroep Professionele Opleidingen
Toegepaste Informatica

Tweede Bachelor Toegepaste Informatica

Integration Project 2

Adam Bouchikhi (r1048050)
Arsalan Borna (r1035349)
Larissa Mbonazo Lusakueno (r1029256)
Mohamed Amin Ahkim (r1038196)
Saartje Cordier (r1037117)

Academiejaar 2025-2026

Mega Data Challenge: Eurovision Voting

Inhoudsopgave

1. Achtergronden (te herzien)

- opdrachtgever: ai assistent

We hebben een project waarbij we een Cloud-gebaseerd systeem willen ontwikkelen die 500 miljoen stemmen voor het Eurovisie Songfestival binnen 60 minuten verwerkt. Dit project heeft twee doelen. Het eerste doel van dit project is dat voor elk land, elk lid het aantal stemmen wordt berekend, waarbij een landcode met een lidnummer wordt toegekend en een timestamp wordt toegevoegd voor het moment van de lokale resultaatberekening. Het tweede doel is dat voor elke stemmer een aantal gegevens worden opgeslagen: het mobiele nummer, de locatie (landcode), het gekozen lidnummer en het tijdstip van de stem.

In dit project ligt de focus niet alleen op het eindresultaat, maar vooral op het ontwikkelen van de technische vaardigheden die nodig zijn om een dataprocessing systeem op te bouwen. Daarom gaan we gebruiken maken van verschillende tools zoals Virtual Machines, Python, virtuele omgevingen, dataprocessing libraries en Docker.

2. Requirements

De belangrijkste eis is het maken van een Proof of Concept (PoC): een werkend model dat laat zien dat het systeem deze enorme hoeveelheid data echt aankan. Het systeem moet stabiel zijn en de gegevens foutloos opslaan. Ook is het een vereiste dat we gebruikmaken van moderne technieken, zoals virtuele omgevingen, om de software overal makkelijk te kunnen draaien.

3. Begrippenlijst

Voordat we de diepte in gaan, hebben we een overzicht gemaakt van de belangrijkste kernwoorden.

- **Virtuele machine**: “Een virtuele machine emuleert een fysieke computer, draait zijn eigen besturingssysteem en apps met gevirtualiseerde middelen. Het is geïsoleerd van het hostsysteem, waardoor gebruikers veilige taken kunnen uitvoeren, zoals het testen van apps of het gebruiken van verschillende besturingssystemen, terwijl ze de fysieke hardware optimaliseren.”
- **Python**: Python is een programmeertaal die begin jaren 90 ontworpen en ontwikkeld werd door Guido van Rossum...

- **Virtuele omgeving - venv**: Een virtuele omgeving in Python is een geïsoleerde omgeving op je computer waar je Python-projecten kunt draaien en testen.
- **Pandas**: Het Python Pandas package is een open source library met data analyse functionaliteit die gebruikt wordt binnen de programmeertaal Python.
- **PySpark**: PySpark is de Python API voor Apache Spark, een open-source, gedistribueerd verwerkingsframework voor big data.
- **Docker container**: Een Docker container is een lichtgewicht, *standalone* uitvoerbare eenheid die alles bevat wat een applicatie nodig heeft om te draaien: van code en runtime tot bibliotheken en instellingen.
- **Proof of Concept**: het proces waarbij bewijs wordt verzameld om de haalbaarheid van een project te bewijzen.

4. Setup

4.1. Virtuele machine

Voor dit project hebben we VirtualBox gebruikt om een virtuele machine te kunnen installeren. Omdat:

- experimenteren veilig uitgevoerd kunnen worden
- software installatie geen problemen veroorzaakt op de host computer
- Het systeem later gemakkelijk kan worden gedeployed

Uit verschillende onderzoeken en marktanalyses blijkt dat Linux de absolute dominante kracht is in de server wereld. Daardoor hebben we beslist om Ubuntu te gebruiken als besturingssysteem. [1]

We hebben vanop de officiële website van Ubuntu een ISO-image gedownload van de Desktop versie.

Nadat we dit bestand hebben gedownload hebben we hem op Oracle VirtualBox een virtuele aangemaakt met deze ISO. Hier hebben we een aantal elementen ingesteld:

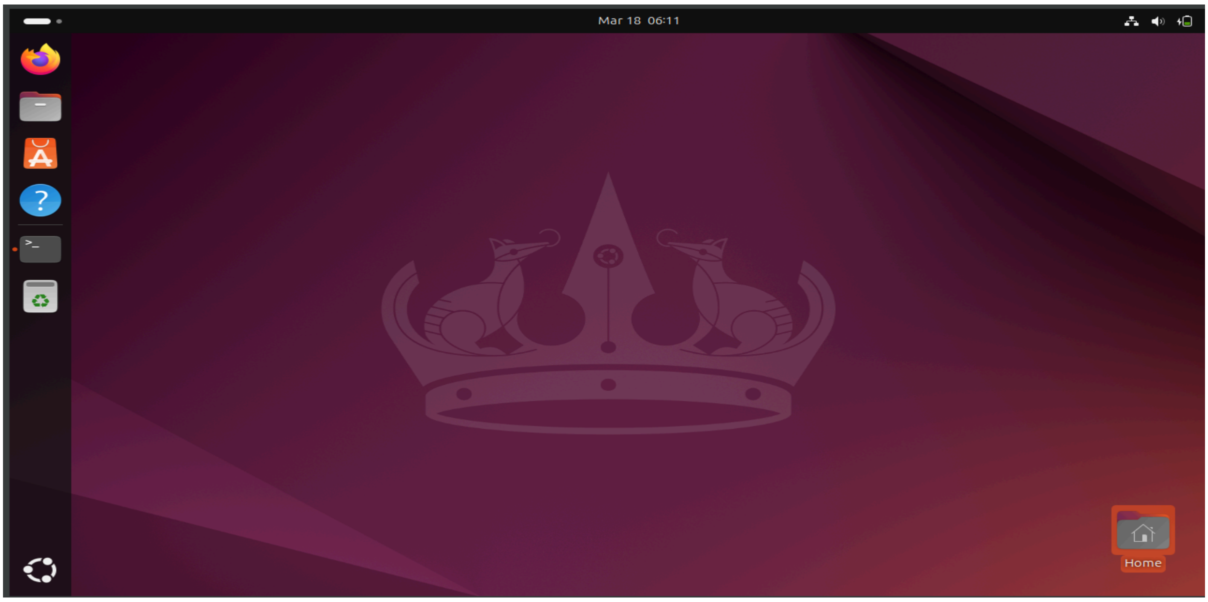
- RAM (geheugen): Hier hebben 8GB RAM gekozen want in ons project draaien verschillende elementen tegelijk (Spark, Docker, Python en Python libraries). Die verbruiken veel geheugen. Als we te weinig RAM hebben dan zou de virtuele machine traag zijn of crashen.
- CPU (processors): We hebben ook voor 4 cores gekozen omdat Spark data parallel verwerkt. Hoe meer cores we hebben hoe sneller data verwerkt en getest wordt.

---

[1] https://commandlinux.com/statistics/linux-adoption-in-fortune-500-companies/#:~:text=Linux%20dominates%20Fortune%20500%20enterprise,90%25%20of%20public%20cloud%20workloads.

- Opslag (disk): Voor ons project hebben we genoeg opslag (25-50 GB) nodig voor het besturingssysteem (Ubuntu), Docker images en Python packages en datasets. Als we te weinig opslag hebben dan zouden de installaties falen of zou Docker geen images kunnen opslaan.

Deze instellingen zorgen ervoor dat de VM voldoende resources heeft voor parallel processing (CPU), in-memory data verwerking (RAM) en containerisatie (Docker).



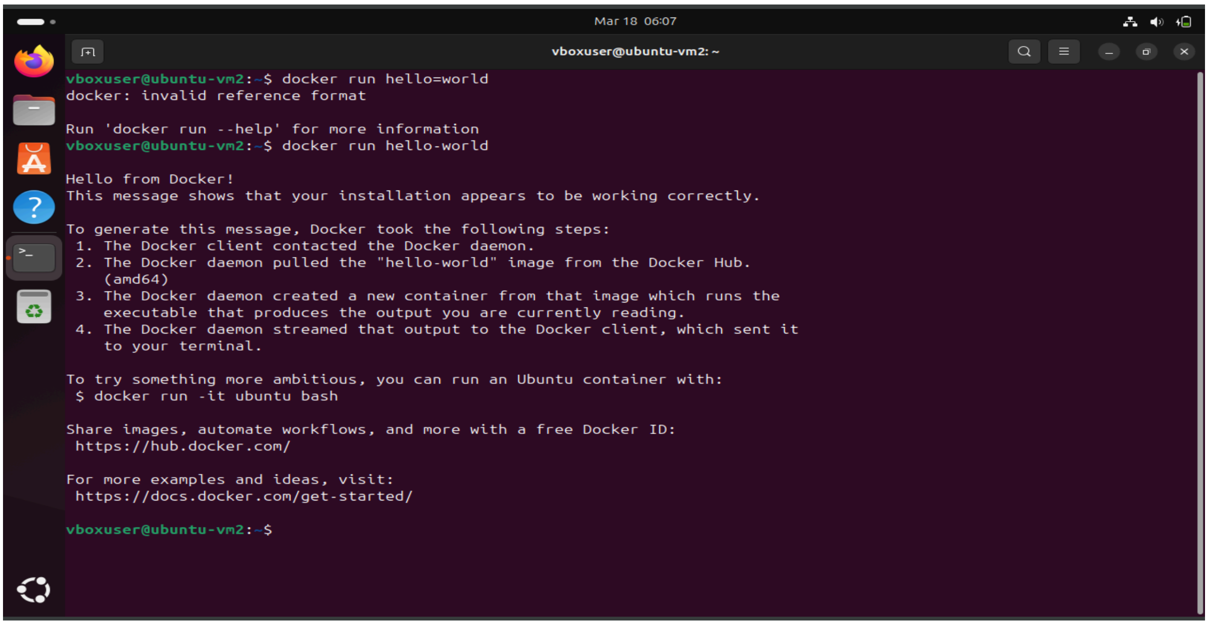
Figuur 1: Ubuntu Virtual Machine omgeving

4.2. Architectuur

4.2.1. Docker

Daarna hebben we ook Docker voor Ubuntu geïnstalleerd. Docker is een technologie waarmee applicaties in containers kunnen draaien. Containers maken het mogelijk om:

1. applicaties snel te deployen 2. services te isoleren 3. systemen schaalbaar te maken



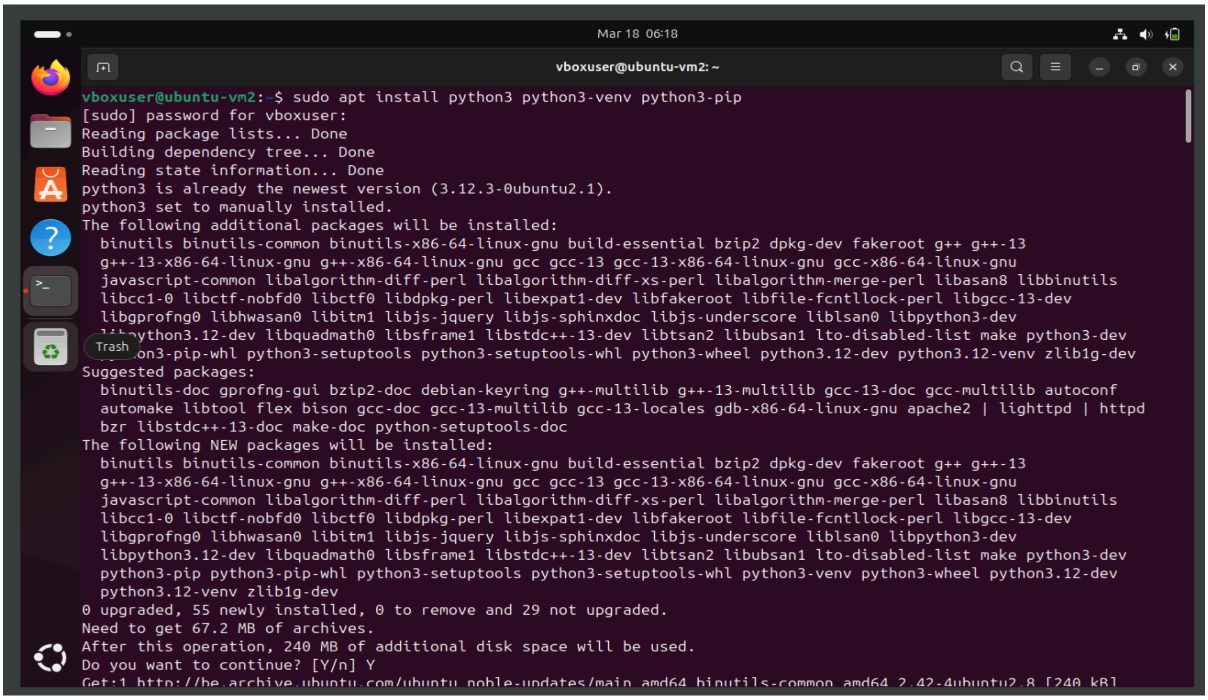
Figuur 2: Docker HELLO-WORLD

4.2.2. Python Virtual Environment (venv)

Om het project volledig te isoleren, is er gebruikgemaakt van een Python Virtual Environment.

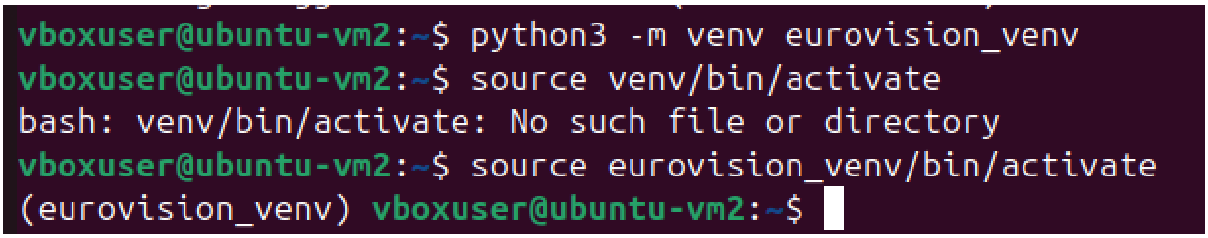
Dit is belangrijk omdat verschillende projecten verschillende library versies kunnen gebruiken. Zonder isolatie kunnen er conflicten ontstaan tussen libraries. Venv levert een stabiele project omgeving. Daarom hebben we gebruik van deze technologie gemaakt.

Eerst hebben we gecontroleerd of Python en de nodige pakketten al aanwezig waren.



Figuur 3: Python installatie

Dan werd de omgeving aangemaakt en geactiveerd via de commando's python3 –m venv eurovision_venv en source eurovision_venv/bin/activate. Het commando python3 –m venv eurovision_venv maakt een nieuwe map aan en daarin zit een aparte python omgeving voor ons project. En het commando eurovision_venv/bin/activate activeert het map dat eerder aangemaakt werd. Op het screenshot hieronder kunnen we zien dat we in de gecreeërd venv omdat ons oorspronkelijke Bash Prompt (vboxuser@ubuntu=vm2) is veranderd naar (eurovision_venv) vboxuser@ubuntu-vm2.



Figuur 4: Python virtual environment (venv)

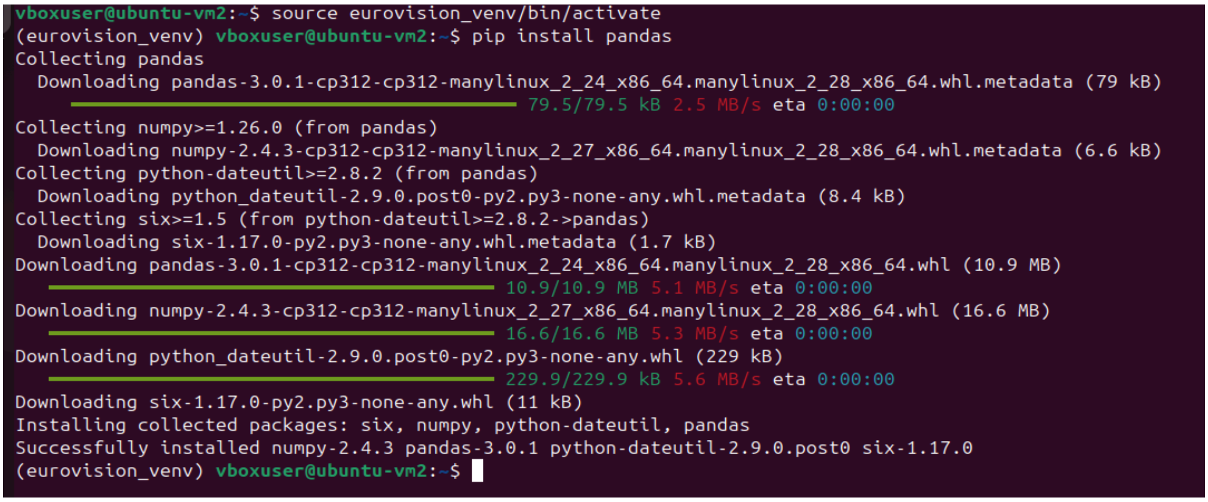
Om te controleren of het werkt hebben we ook het volgend commando: which python in onze terminal terminal getypt. En de output had ons getoond dat python uit de venv kwam en niet uit het systeem.

4.2.3. Pandas

Binnen de geactiveerde venv is Pandas geïnstalleerd via pip install pandas. Dit is belangrijk want Pandas is een Python library die gebruikt wordt voor:

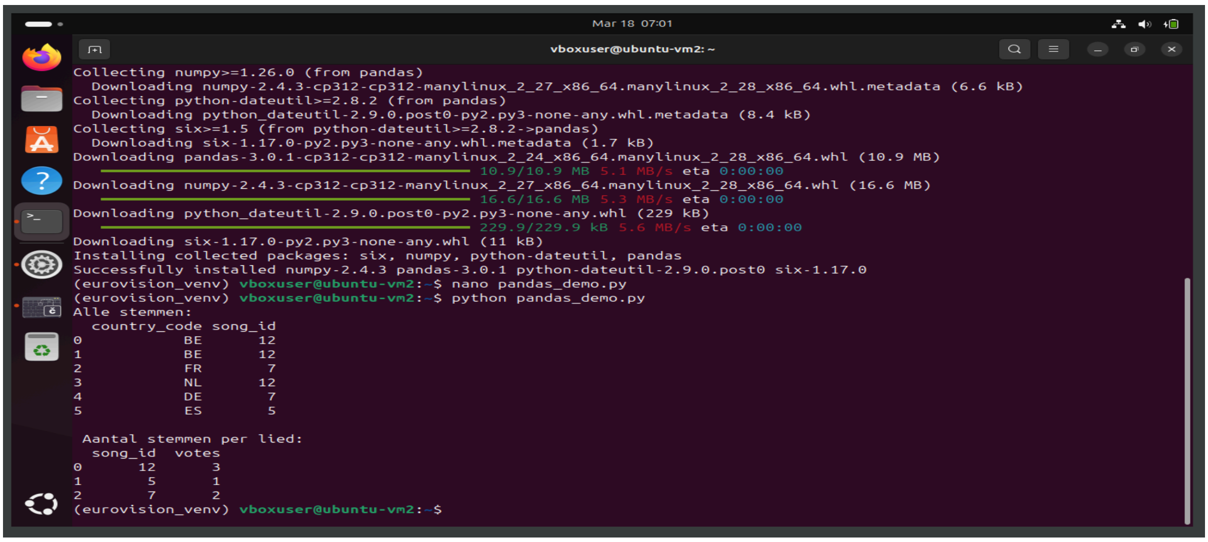
- data analyse
- data cleaning
- data verwerking

Met Pandas kunnen we datasets makkelijk laden, aanpassen en analyseren.



Figuur 5: Pandas installatie

Daarna hebben we een demo van Pandas uitgevoerd om te verifiëren dat de library goed geïnstalleerd is.



Figuur 6

4.2.4. PySpark

Aangezien het Eurovision-project grote hoeveelheden data in korte tijd moet verwerken, is Apache Spark de gekozen technologie voor de schaalbare verwerking.

De Spark-integratie voor Python is geïnstalleerd met pip install pyspark.

```
vboxuser@ubuntu-vm2:~$ source eurovision_venv/bin/activate
(eurovision_venv) vboxuser@ubuntu-vm2:~$ pip install pyspark
Collecting pyspark
  Downloading pyspark-4.1.1.tar.gz (455.4 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 455.4/455.4 MB 2.0 MB/s eta 0:00:00
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Collecting py4j<0.10.9.10,>=0.10.9.7 (from pyspark)
  Downloading py4j-0.10.9.9-py2.py3-none-any.whl.metadata (1.3 kB)
Downloading py4j-0.10.9.9-py2.py3-none-any.whl (203 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 203.0/203.0 kB 6.3 MB/s eta 0:00:00
Building wheels for collected packages: pyspark
  Building wheel for pyspark (pyproject.toml) ... done
  Created wheel for pyspark: filename=pyspark-4.1.1-py2.py3-none-any.whl size=456008704 sha256=da3d678f078f96507c9e9a05d4350aaf2f9dfb6c16abb4b95b6c4d36aa2068e2
  Stored in directory: /home/vboxuser/.cache/pip/wheels/f4/ca/ea/203f40b3e935bbf99bee851c2f4a87d22996ab8212d367ce58
Successfully built pyspark
Installing collected packages: py4j, pyspark
Successfully installed py4j-0.10.9.9 pyspark-4.1.1
(eurovision_venv) vboxuser@ubuntu-vm2:~$
```

Figuur 7: Payspark instalatie

Vervolgens hebben we een demo-script uitgevoerd met PySpark om de installatie te verifiëren.

Hieronder staat de script dat we als test gebruikt hebben:

from pyspark.sql import SparkSession

from pyspark.sql.functions import count

spark = SparkSession.builder.appName("EurovisionPoC").getOrCreate()

data = [

("BE","FR"),("BE","NL"),("NL","FR"),("FR","BE"),("BE","FR"),

("NL","BE"),("DE","FR"),("DE","NL"),("FR","NL"),("NL","DE"),

("BE","DE"),("FR","BE"),("DE","BE"),("NL","FR"),("BE","FR"),

("FR","NL"),("DE","FR"),("NL","BE"),("BE","NL"),("FR","DE")

]

df = spark.createDataFrame(data, ["from_country","to_country"])

print("=== INPUT DATA ==="); df.show()

print("=== STEMMEN PER LAND (LIED) ===")

votes_per_country = df.groupBy("to_country").agg(count("*").alias("votes"))

votes_per_country.show()

print("=== WINNAAR ===")

votes_per_country.orderBy("votes", ascending=False).show(1)

print("=== STEMGEDRAG (VAN → NAAR) ===")

***df.groupBy("from_country","to_country") ***

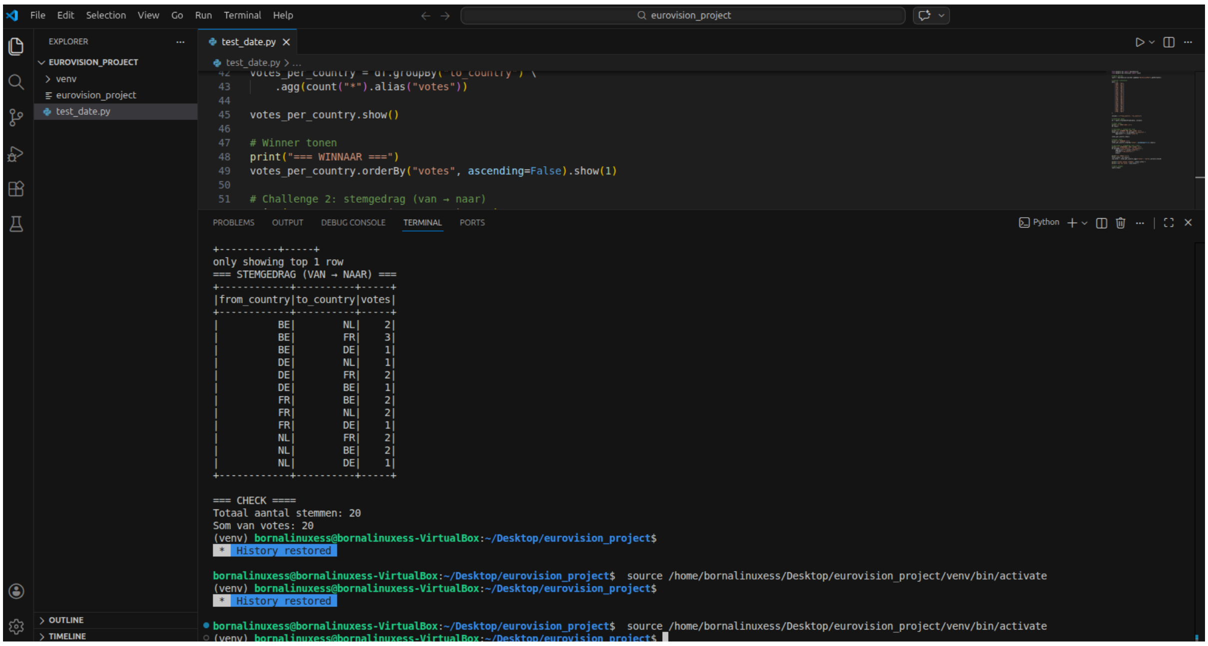
***.agg(count("*").alias("votes")) ***

***.orderBy("from_country") ***

.show()

spark.stop()

Output:



Figuur 8: Output van het systeem met verwerking van stem data en bepaling van de winnaar

4.2.5. Visual Studio Code

We hebben Visual Studio Code vanop de officiële website gedownload op onze virtuele machine.

Hier laat de terminal zien dat de benodigde tools (wget en gpg) al aanwezig zijn.

```
vboxuser@ubuntu-vm2:~$ sudo apt-get install wget gpg && wget -qO- https://packages.microsoft.com/keys/microsoft.asc | gp
g --dearmor > microsoft.gpg && sudo install -D -o root -g root -m 644 microsoft.gpg /usr/share/keyrings/microsoft.gpg &&
 rm -f microsoft.gpg
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
wget is already the newest version (1.21.4-1ubuntu4.1).
gpg is already the newest version (2.4.4-2ubuntu17.4).
0 upgraded, 0 newly installed, 0 to remove and 40 not upgraded.
vboxuser@ubuntu-vm2:~$
```

Figuur 9

```
vboxuser@ubuntu-vm2:~$ sudo nano /etc/apt/sources.list.d/vscode.sources
```

Figuur 10

Hieronder staat de inhoud van het bestand:

Types: deb

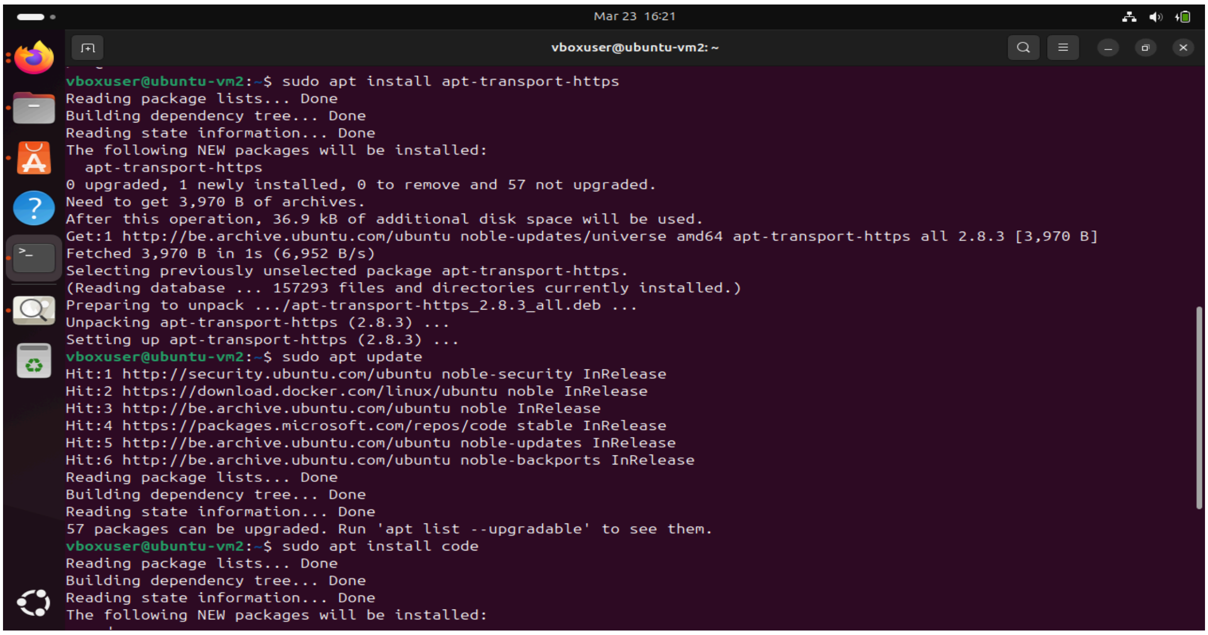
URIs: https://packages.microsoft.com/repos/code

Suites: stable

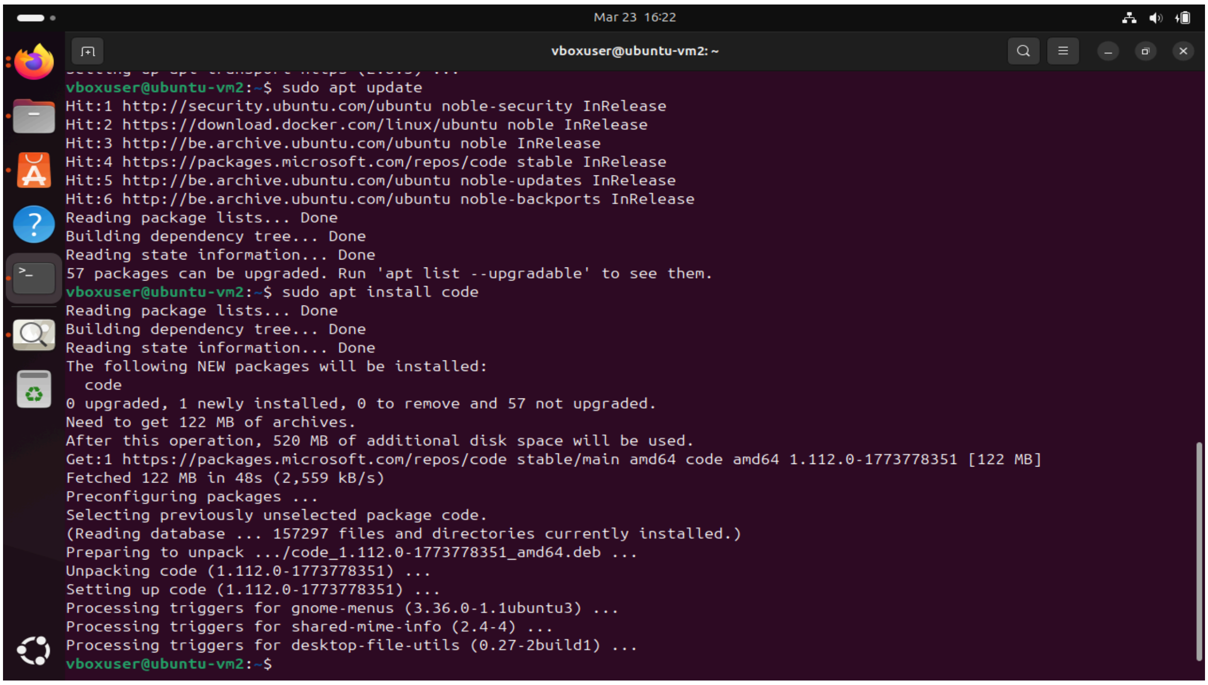
Components: main

Architectures: amd64,arm64,armhf

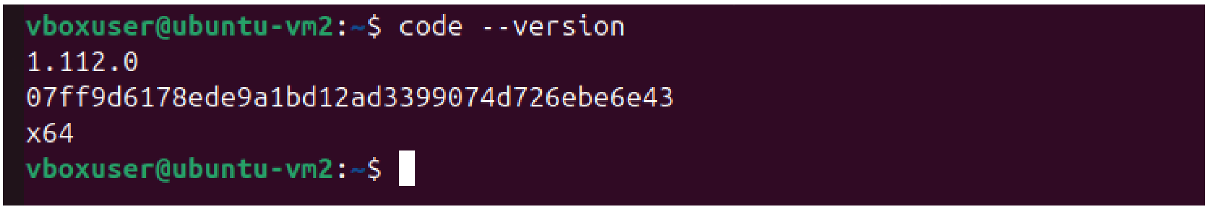
Signed-By: /usr/share/keyrings/microsoft.gpg



Figuur 11

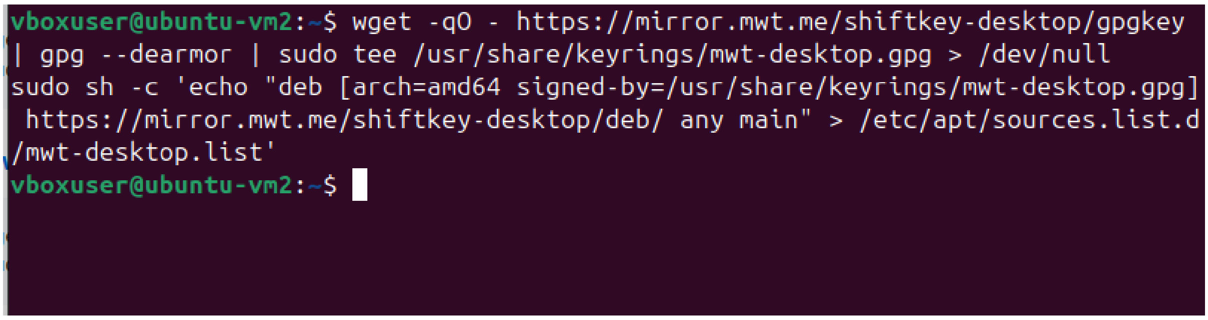


Figuur 12

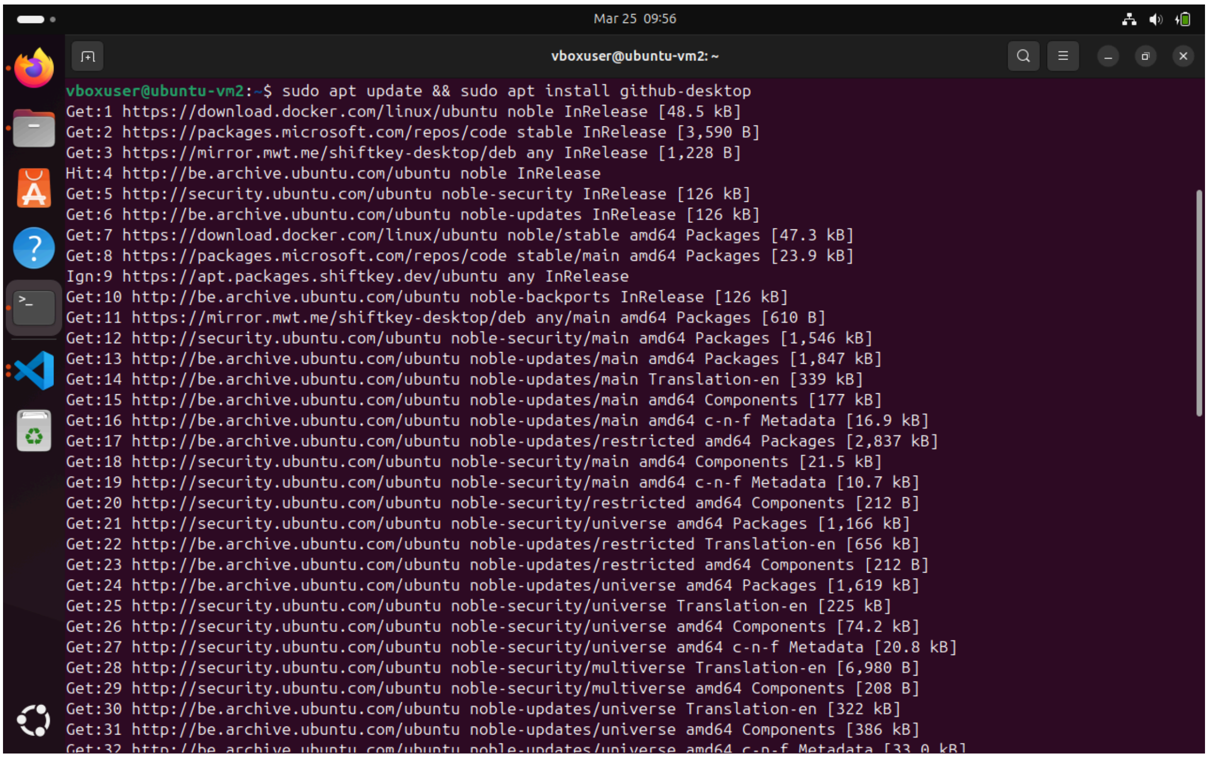


Figuur 13

4.2.6. Github Desktop



Figuur 14



Figuur 15

5. Planning

April:

- Analyse en voorbereiding afronden
- Installatie en setup
- Eerste PoC maken

Mei:

- PoC uitbreiden (meer data)
- Testing en validatie
- Documentatie schrijven

Juni:

- Document afwerken
- Presentatie voorbereiden
- Demo testen

We hebben een Proof of Concept gebouwd met Python en PySpark. Stemmen kunnen verwerkt, geanalyseerd en gebruikt worden om een winnaar te bepalen. Momenteel ligt de focus op:

- documentatie afronden
- presentatie voorbereiden
- demo finaliseren

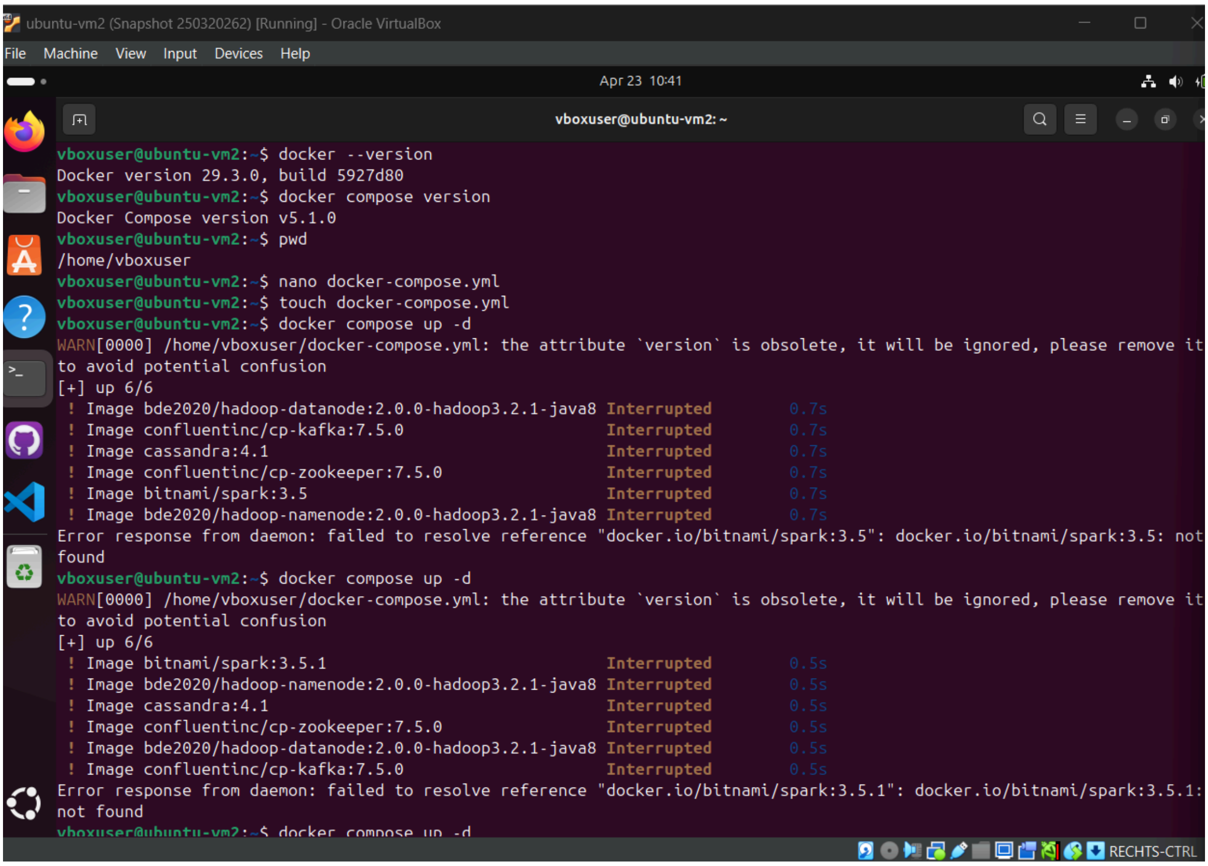
De planning loopt tot juni, met verdere testen en optimalisatie. Tot nu toe is er gewerkt aan de basis van het project.

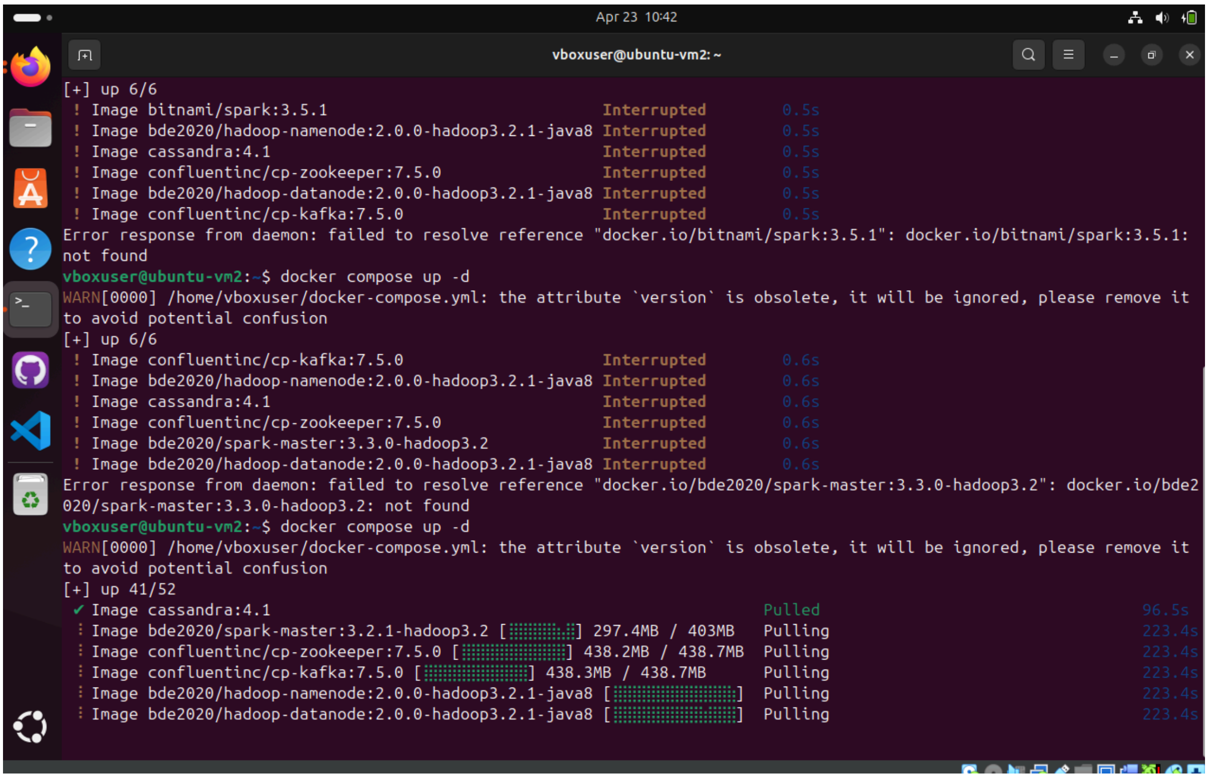
De requirements zijn bepaald en er is een plan van aanpak opgesteld met de belangrijkste taken. Daarnaast is er onderzoek uitgevoerd naar de gebruikte technologieën en zijn de tools geconfigureerd, zoals Python, PySpark en de Virtual Machine.

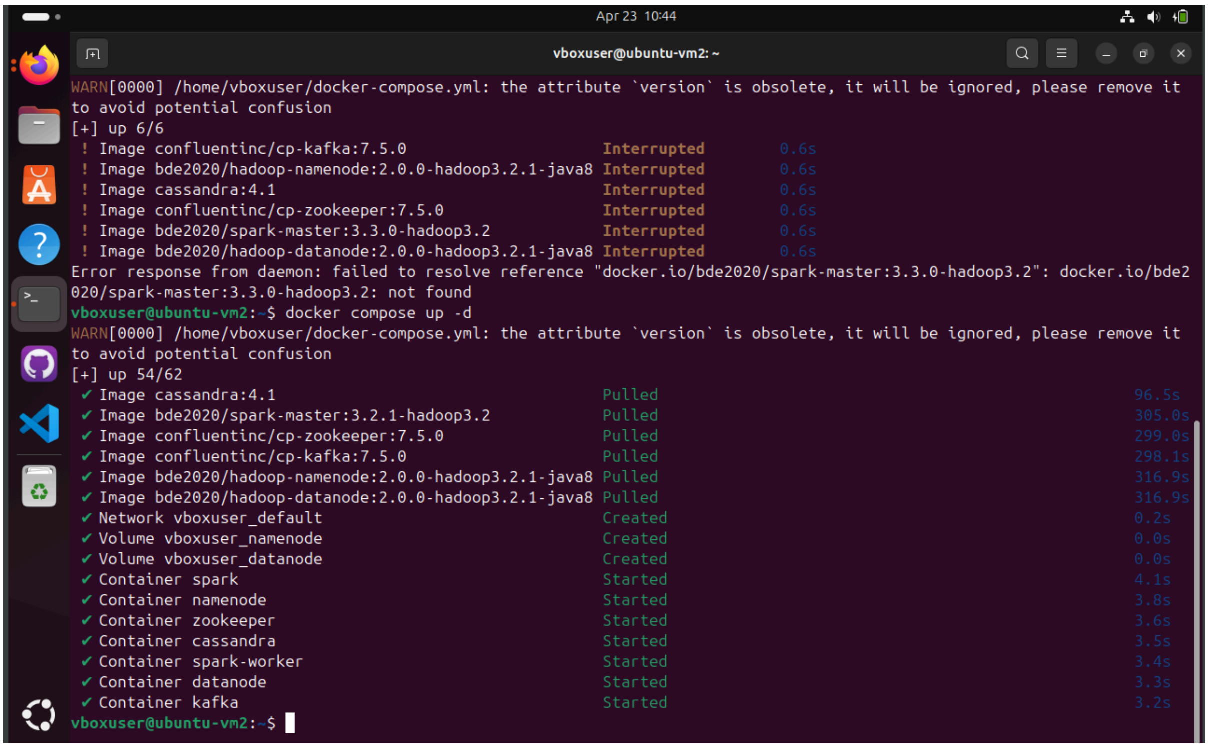
Vervolgens is een eerste planning opgesteld en een eenvoudige architectuur uitgewerkt van hoe het systeem functioneert. Ook is de documentatie opgesteld en een Proof of Concept gebouwd waarin stemmen worden verwerkt, geanalyseerd en de winnaar wordt bepaald.

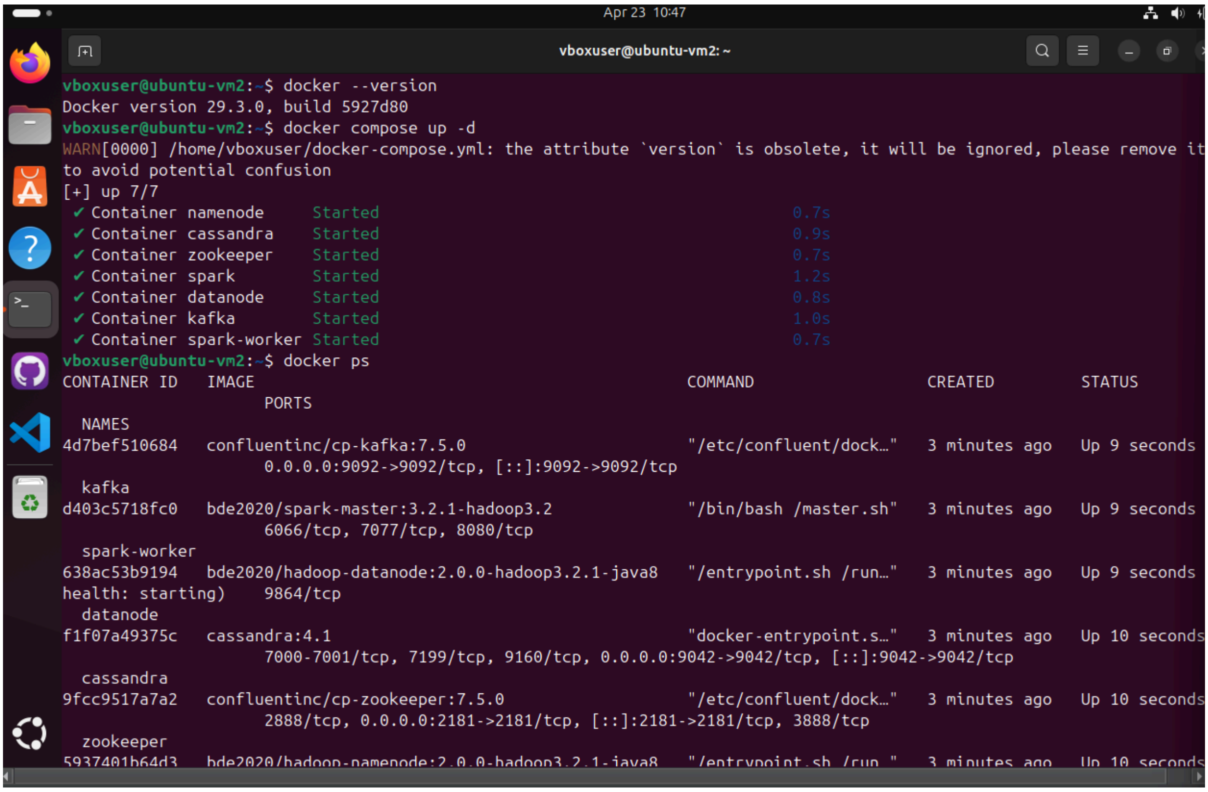
Tot slot is de oplossing getest om te controleren of alles correct werkt.

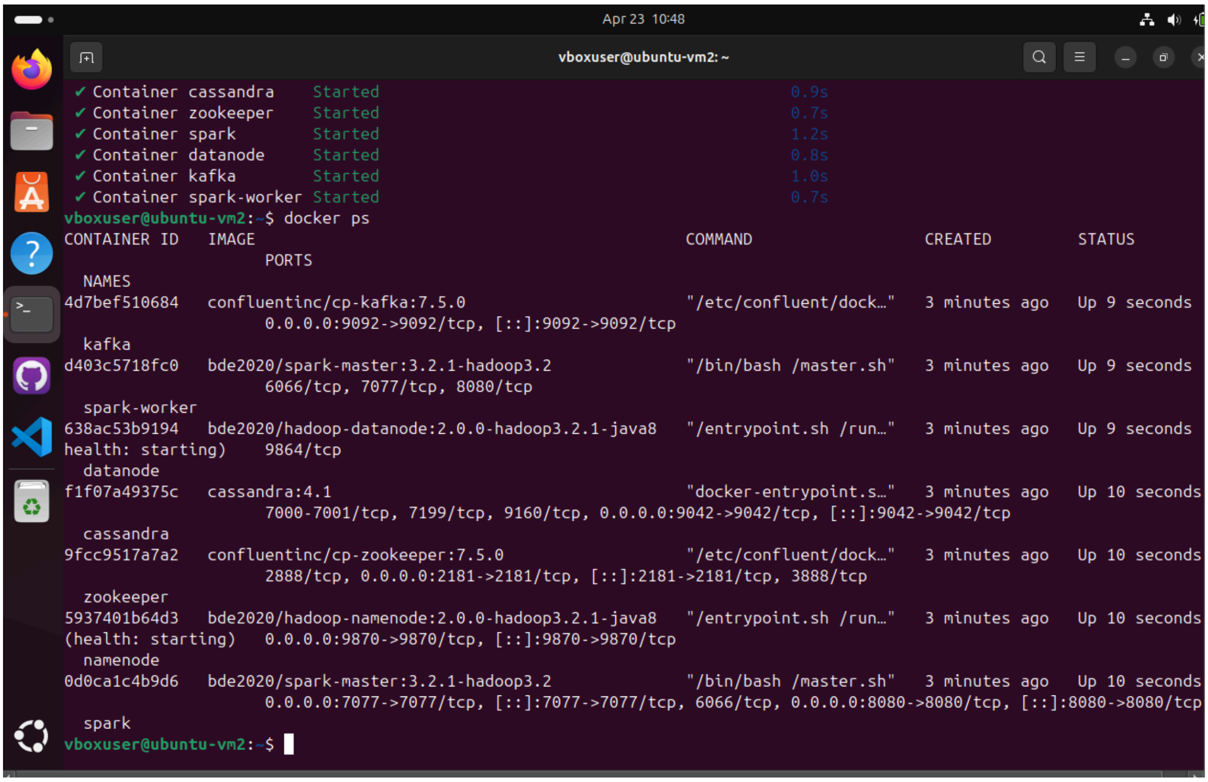
In de volgende fase ligt de nadruk op het verder verbeteren van de oplossing, het uitbreiden van de testen en het voorbereiden van de presentatie.



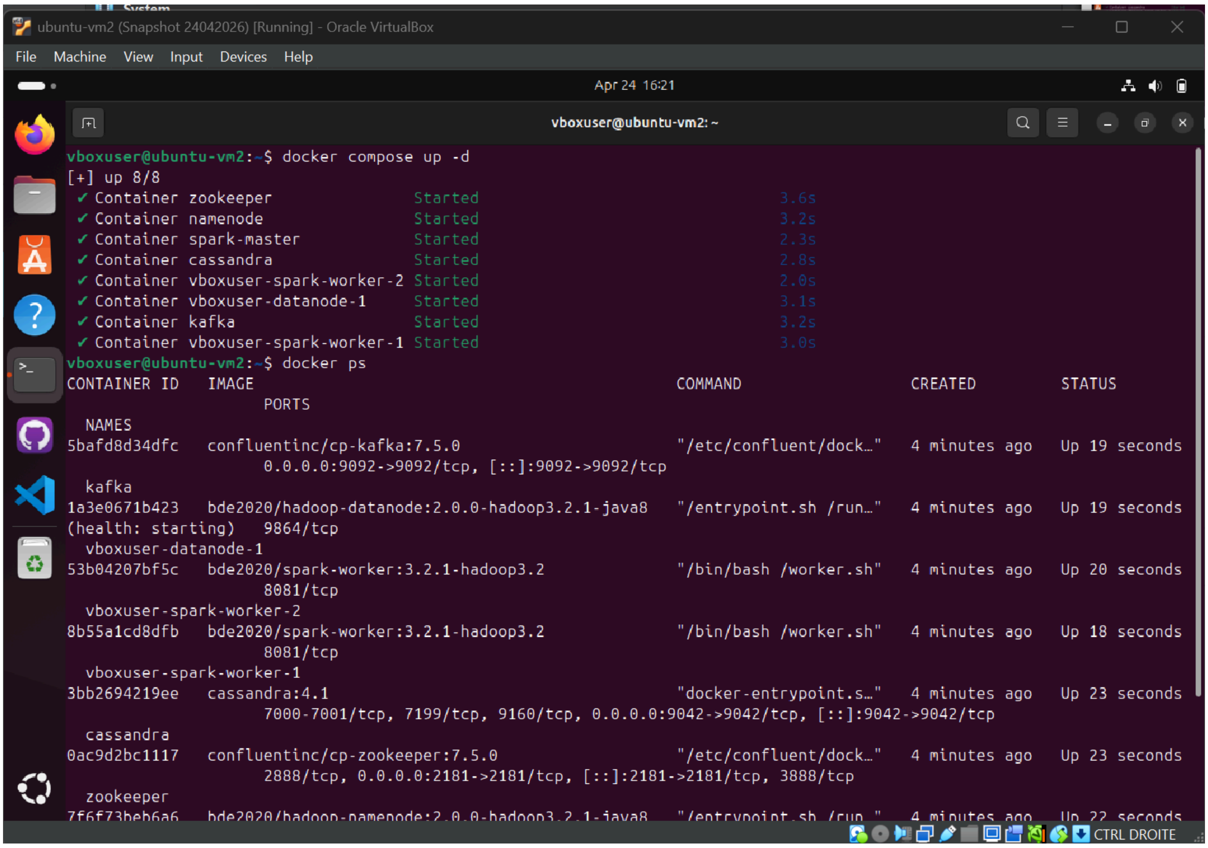

Apr 23 10:42
vboxuser@ubuntu-vm2: ~
[+] up 6/6
! Image bitnami/spark:3.5.1 Interrupted 0.5s
! Image bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8 Interrupted 0.5s
! Image cassandra:4.1 Interrupted 0.5s
! Image confluentinc/cp-zookeeper:7.5.0 Interrupted 0.5s
! Image bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8 Interrupted 0.5s
! Image confluentinc/cp-kafka:7.5.0 Interrupted 0.5s
Error response from daemon: failed to resolve reference "docker.io/bitnami/spark:3.5.1": docker.io/bitnami/spark:3.5.1: not found
vboxuser@ubuntu-vm2:~$ docker compose up -d
WARN[0000] /home/vboxuser/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 6/6
! Image confluentinc/cp-kafka:7.5.0 Interrupted 0.6s
! Image bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8 Interrupted 0.6s
! Image cassandra:4.1 Interrupted 0.6s
! Image confluentinc/cp-zookeeper:7.5.0 Interrupted 0.6s
! Image bde2020/spark-master:3.3.0-hadoop3.2 Interrupted 0.6s
! Image bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8 Interrupted 0.6s
Error response from daemon: failed to resolve reference "docker.io/bde2020/spark-master:3.3.0-hadoop3.2": docker.io/bde2020/spark-master:3.3.0-hadoop3.2: not found
vboxuser@ubuntu-vm2:~$ docker compose up -d
WARN[0000] /home/vboxuser/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 41/52
✔ Image cassandra:4.1 Pulled 96.5s
Image bde2020/spark-master:3.2.1-hadoop3.2 [] 297.4MB / 403MB Pulling 223.4s
Image confluentinc/cp-zookeeper:7.5.0 [] 438.2MB / 438.7MB Pulling 223.4s
Image confluentinc/cp-kafka:7.5.0 [] 438.3MB / 438.7MB Pulling 223.4s
Image bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8 [] Pulling 223.4s
Image bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8 [] Pulling 223.4s


Apr 23 10:44
vboxuser@ubuntu-vm2: ~
WARN[0000] /home/vboxuser/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 6/6
! Image confluentinc/cp-kafka:7.5.0 Interrupted 0.6s
! Image bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8 Interrupted 0.6s
! Image cassandra:4.1 Interrupted 0.6s
! Image confluentinc/cp-zookeeper:7.5.0 Interrupted 0.6s
! Image bde2020/spark-master:3.3.0-hadoop3.2 Interrupted 0.6s
! Image bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8 Interrupted 0.6s
Error response from daemon: failed to resolve reference "docker.io/bde2020/spark-master:3.3.0-hadoop3.2": docker.io/bde2020/spark-master:3.3.0-hadoop3.2: not found
vboxuser@ubuntu-vm2:~$ docker compose up -d
WARN[0000] /home/vboxuser/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 54/62
✔ Image cassandra:4.1 Pulled 96.5s
✔ Image bde2020/spark-master:3.2.1-hadoop3.2 Pulled 305.0s
✔ Image confluentinc/cp-zookeeper:7.5.0 Pulled 299.0s
✔ Image confluentinc/cp-kafka:7.5.0 Pulled 298.1s
✔ Image bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8 Pulled 316.9s
✔ Image bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8 Pulled 316.9s
✔ Network vboxuser_default Created 0.2s
✔ Volume vboxuser_namenode Created 0.0s
✔ Volume vboxuser_datanode Created 0.0s
✔ Container spark Started 4.1s
✔ Container namenode Started 3.8s
✔ Container zookeeper Started 3.6s
✔ Container cassandra Started 3.5s
✔ Container spark-worker Started 3.4s
✔ Container datanode Started 3.3s
✔ Container kafka Started 3.2s
vboxuser@ubuntu-vm2:~$


Apr 23 10:47
vboxuser@ubuntu-vm2: ~
vboxuser@ubuntu-vm2:~$ docker --version
Docker version 29.3.0, build 5927d80
vboxuser@ubuntu-vm2:~$ docker compose up -d
WARN[0000] /home/vboxuser/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 7/7
✔ Container namenode Started 0.7s
✔ Container cassandra Started 0.9s
✔ Container zookeeper Started 0.7s
✔ Container spark Started 1.2s
✔ Container datanode Started 0.8s
✔ Container kafka Started 1.0s
✔ Container spark-worker Started 0.7s
vboxuser@ubuntu-vm2:~$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
4d7bef510684 confluentinc/cp-kafka:7.5.0 "/etc/confluent/dock…" 3 minutes ago Up 9 seconds 0.0.0.0:9092->9092/tcp, [::]:9092->9092/tcp kafka
d403c5718fc0 bde2020/spark-master:3.2.1-hadoop3.2 "/bin/bash /master.sh" 3 minutes ago Up 9 seconds 6066/tcp, 7077/tcp, 8080/tcp spark-worker
638ac53b9194 bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8 "/entrypoint.sh /run…" 3 minutes ago Up 9 seconds health: starting) 9864/tcp datanode
f1f07a49375c cassandra:4.1 "docker-entrypoint.s…" 3 minutes ago Up 10 seconds 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, [::]:9042->9042/tcp cassandra
9fcc9517a7a2 confluentinc/cp-zookeeper:7.5.0 "/etc/confluent/dock…" 3 minutes ago Up 10 seconds 2888/tcp, 0.0.0.0:2181->2181/tcp, [::]:2181->2181/tcp, 3888/tcp zookeeper

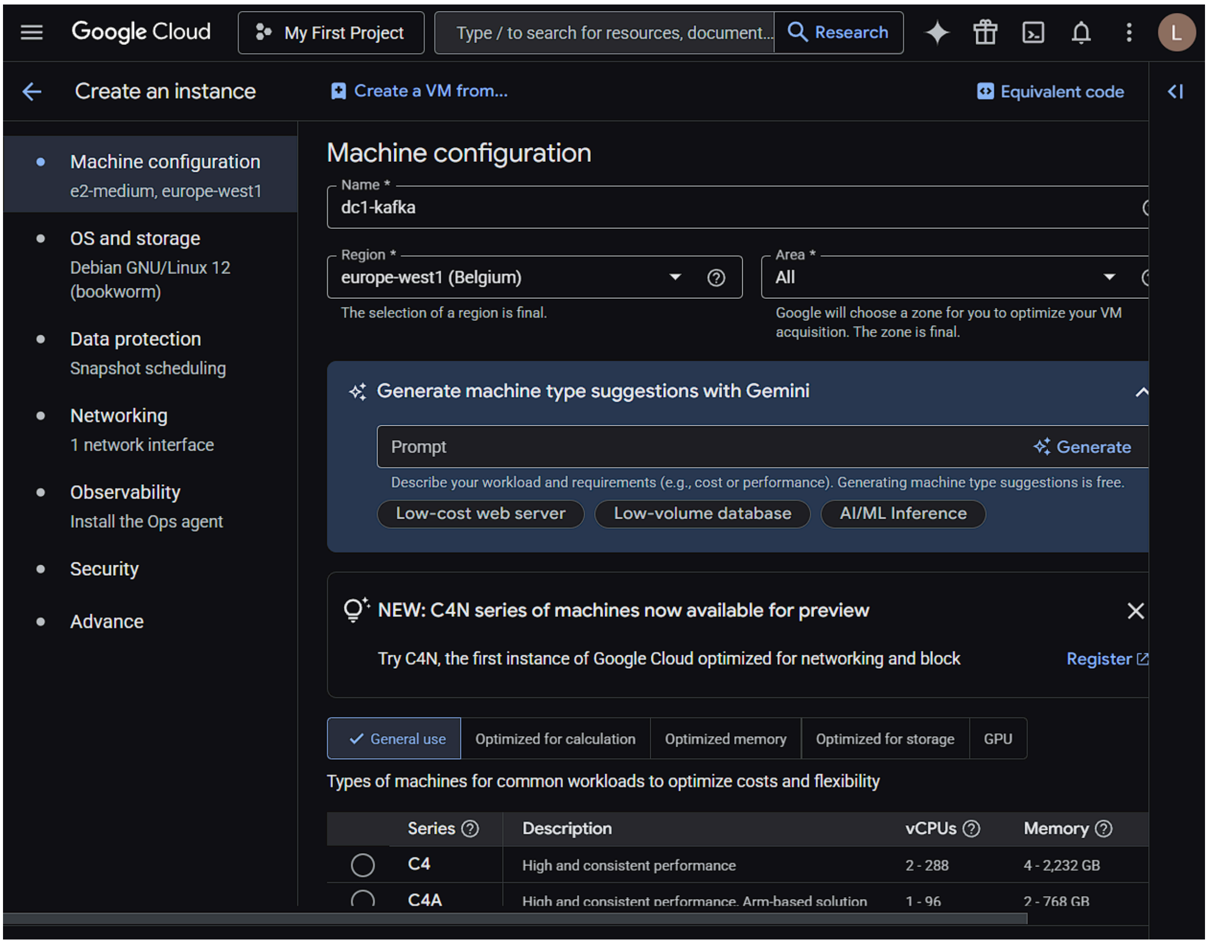


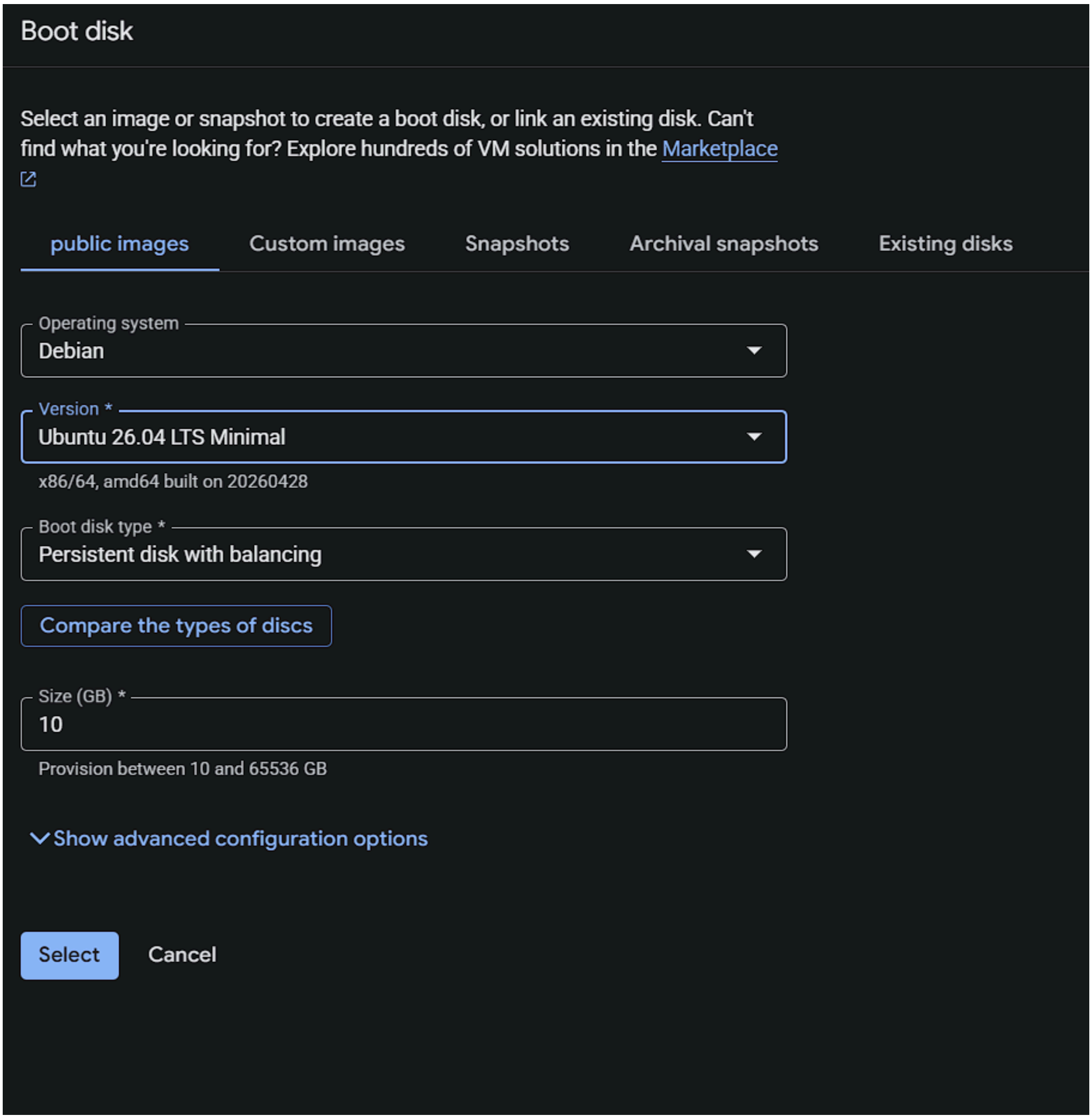
na troubleshooting (ook script)

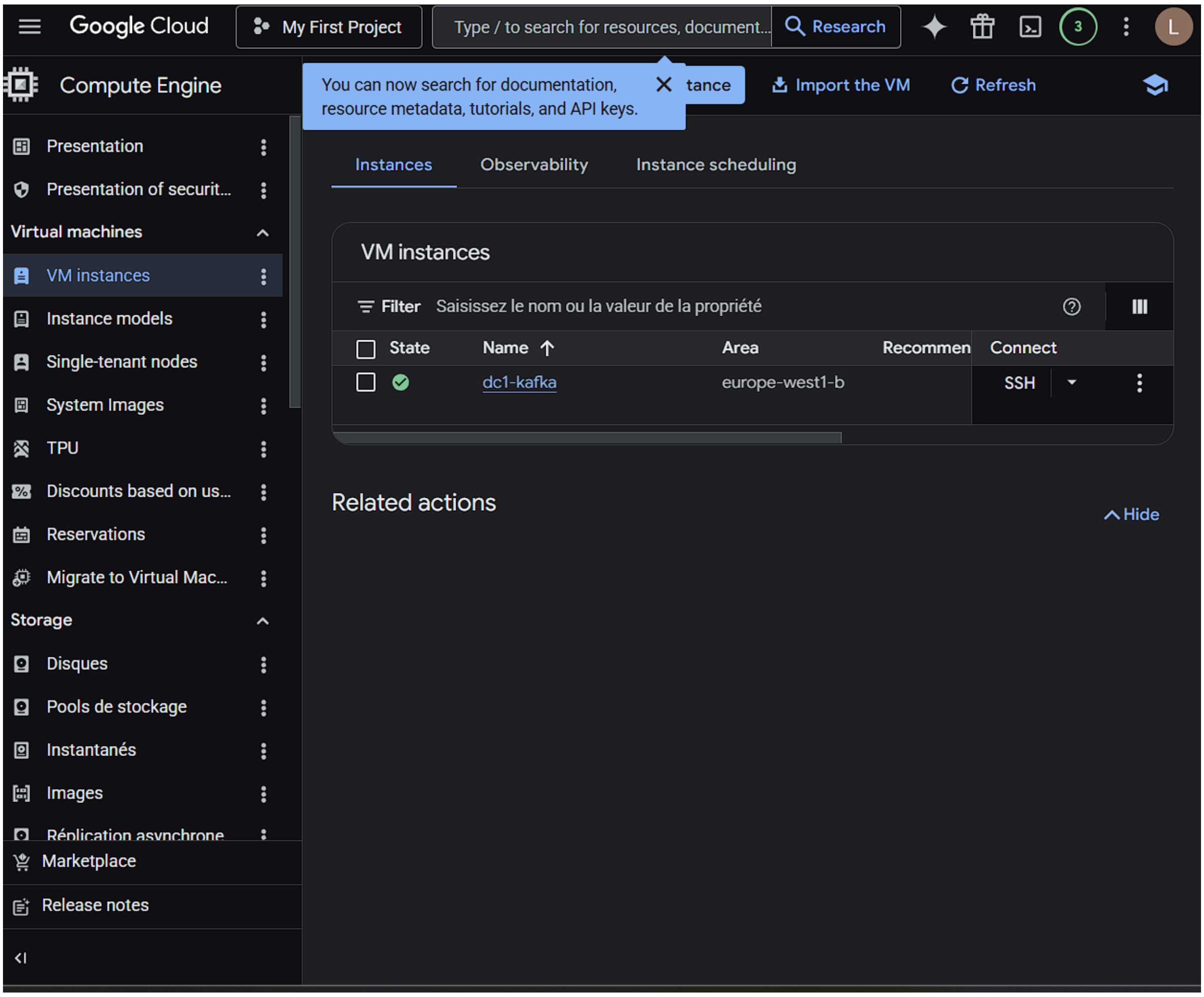


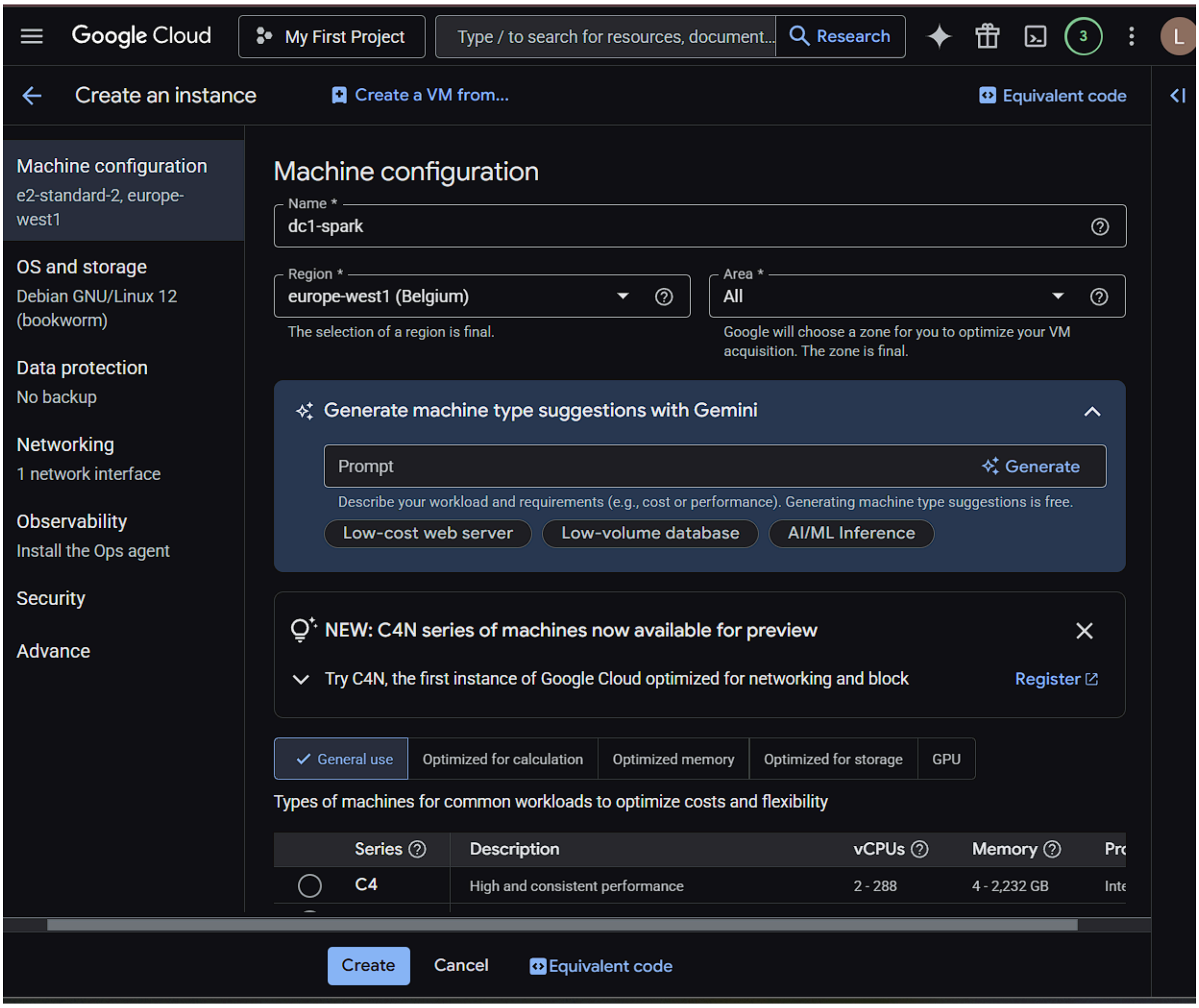
docker compose up -d

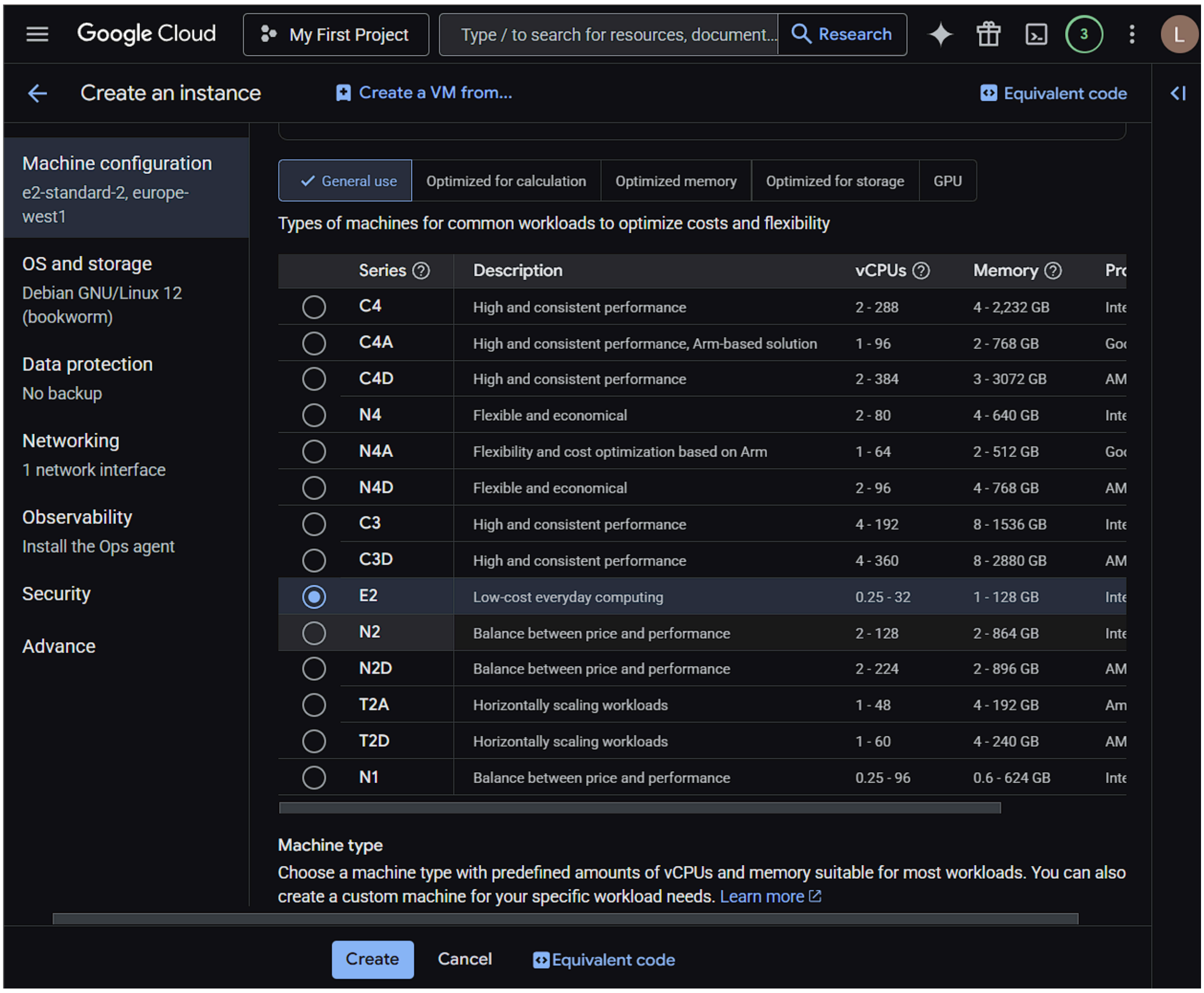
docker ps

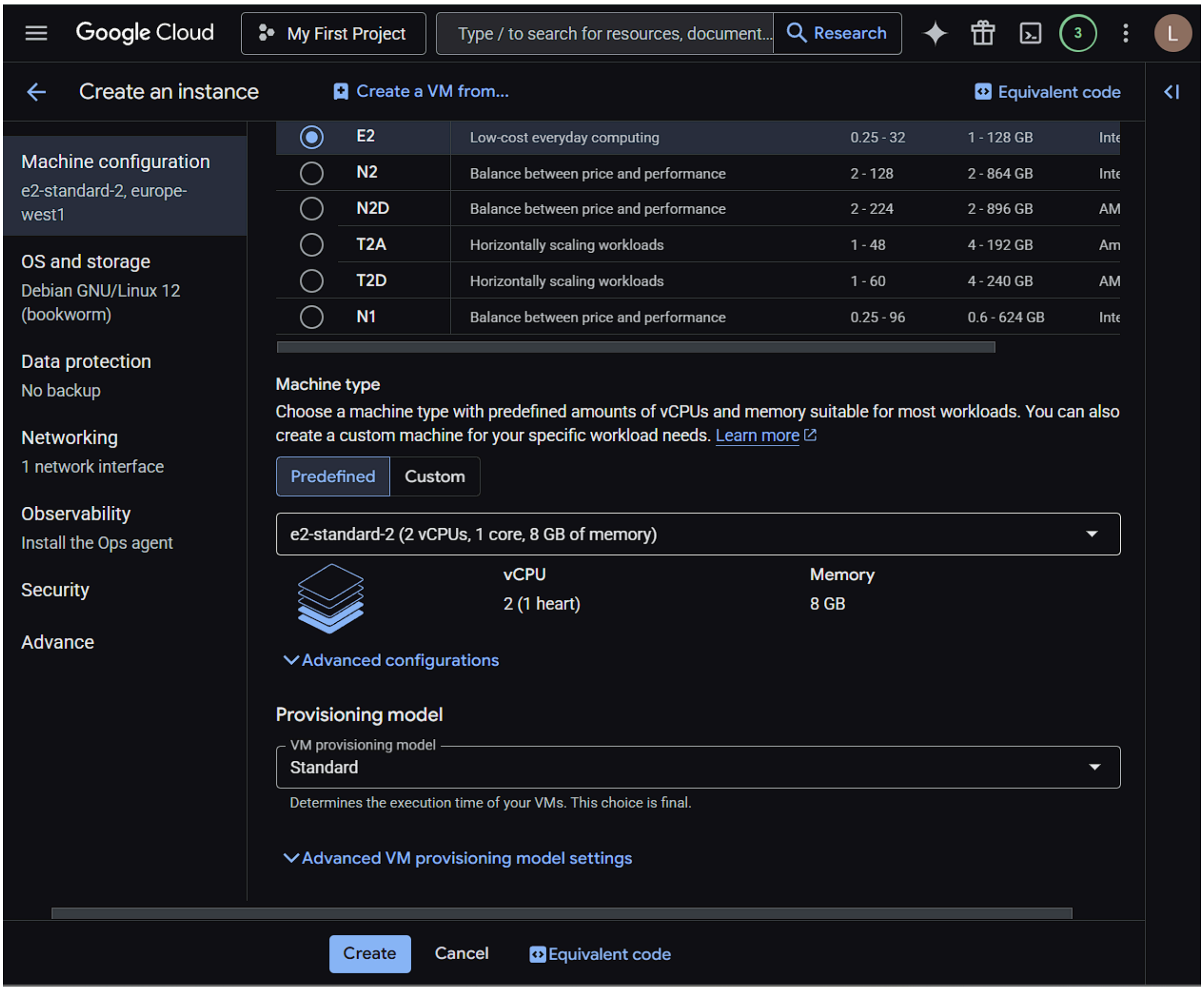

Google Cloud
My First Project
Type / to search for resources, document...
Research
Create an instance
Create a VM from...
Equivalent code
Machine configuration
e2-medium, europe-west1
OS and storage
Debian GNU/Linux 12 (bookworm)
Data protection
Snapshot scheduling
Networking
1 network interface
Observability
Install the Ops agent
Security
Advance
Machine configuration
Name *
dc1-kafka
Region *
europe-west1 (Belgium)
The selection of a region is final.
Area *
All
Google will choose a zone for you to optimize your VM acquisition. The zone is final.
Generate machine type suggestions with Gemini
Prompt
Generate
Describe your workload and requirements (e.g., cost or performance). Generating machine type suggestions is free.
Low-cost web server
Low-volume database
AI/ML Inference
NEW: C4N series of machines now available for preview
Try C4N, the first instance of Google Cloud optimized for networking and block
Register
General use
Optimized for calculation
Optimized memory
Optimized for storage
GPU
Types of machines for common workloads to optimize costs and flexibility
Series
Description
vCPUs
Memory
C4
High and consistent performance
2 - 288
4 - 2,232 GB
C4A
High and consistent performance. Arm-based solution
1 - 96
2 - 768 GB

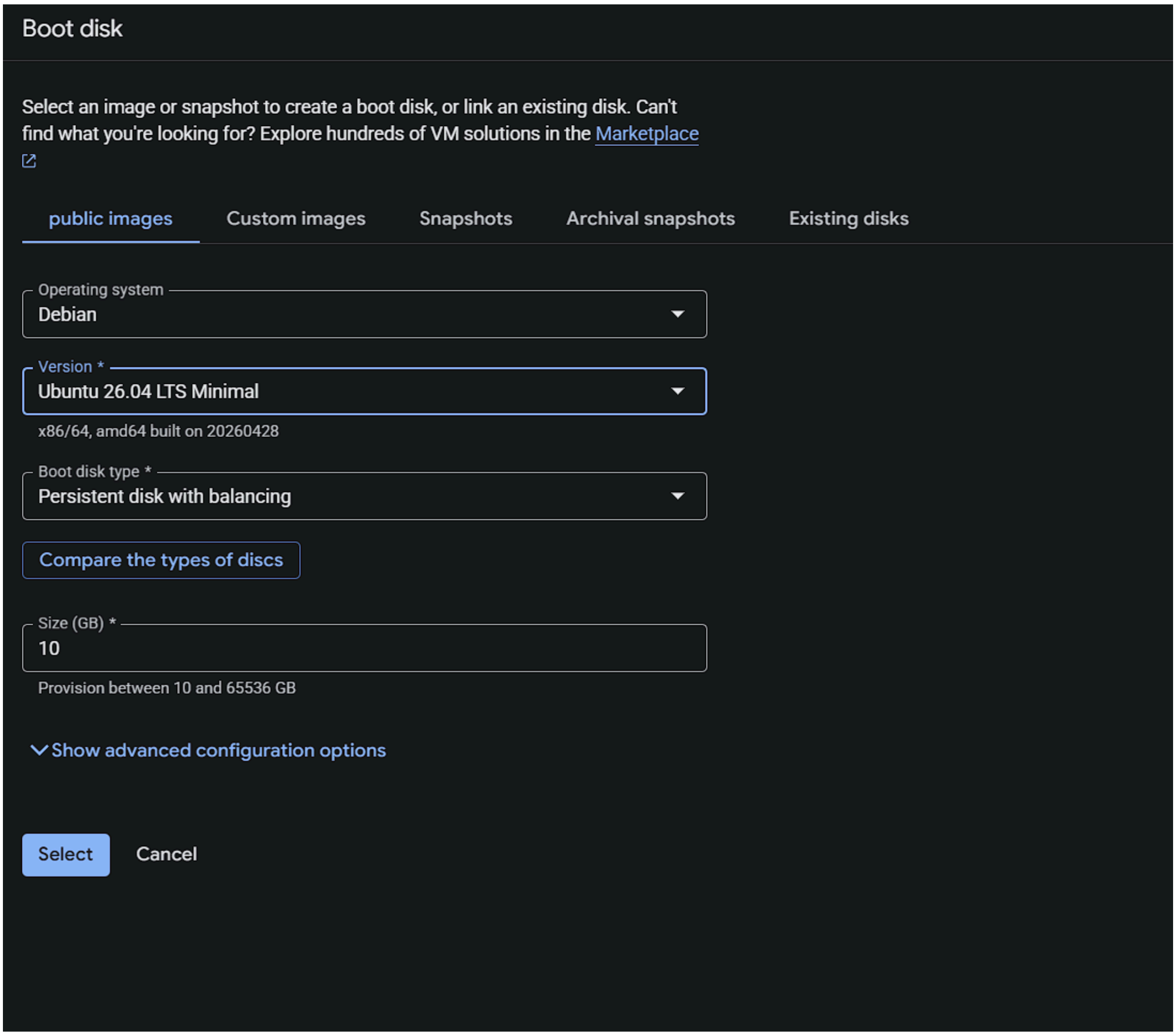

Boot disk
Select an image or snapshot to create a boot disk, or link an existing disk. Can't find what you're looking for? Explore hundreds of VM solutions in the Marketplace
public images
Custom images
Snapshots
Archival snapshots
Existing disks
Operating system
Debian
Version *
Ubuntu 26.04 LTS Minimal
x86/64, amd64 built on 20260428
Boot disk type *
Persistent disk with balancing
Compare the types of discs
Size (GB) *
10
Provision between 10 and 65536 GB
Show advanced configuration options
Select
Cancel

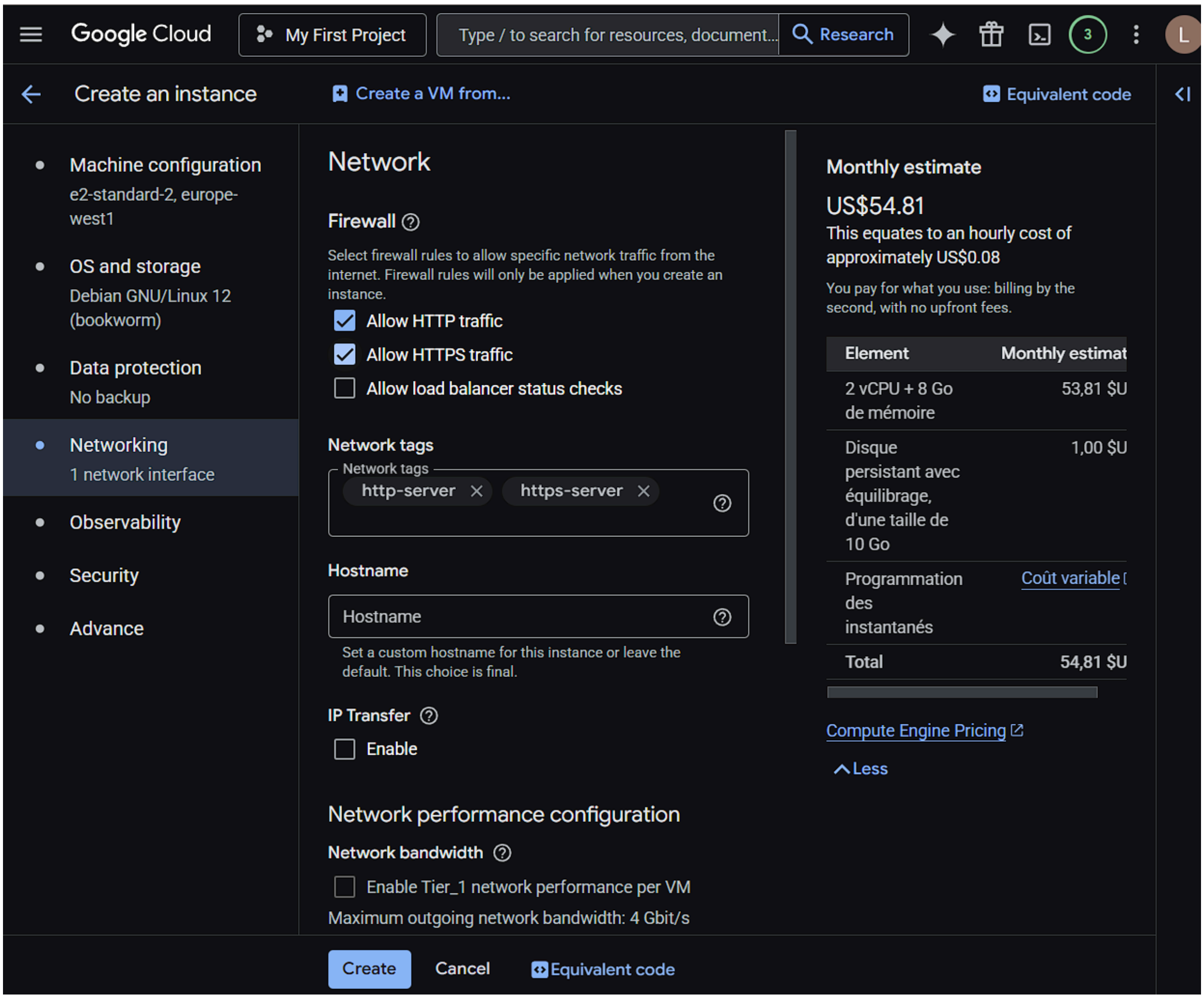

Google Cloud
My First Project
Type / to search for resources, document...
Research
Compute Engine
You can now search for documentation, resource metadata, tutorials, and API keys.
tance
Import the VM
Refresh
Presentation
Presentation of securit...
Virtual machines
VM instances
Instance models
Single-tenant nodes
System Images
TPU
Discounts based on us...
Reservations
Migrate to Virtual Mac...
Storage
Disques
Pools de stockage
Instantanés
Images
Réplication asynchrone
Marketplace
Release notes
Instances
Observability
Instance scheduling
VM instances
Filter
Saisissez le nom ou la valeur de la propriété
State
Name
Area
Recommen
Connect
dc1-kafka
europe-west1-b
SSH
Related actions
Hide

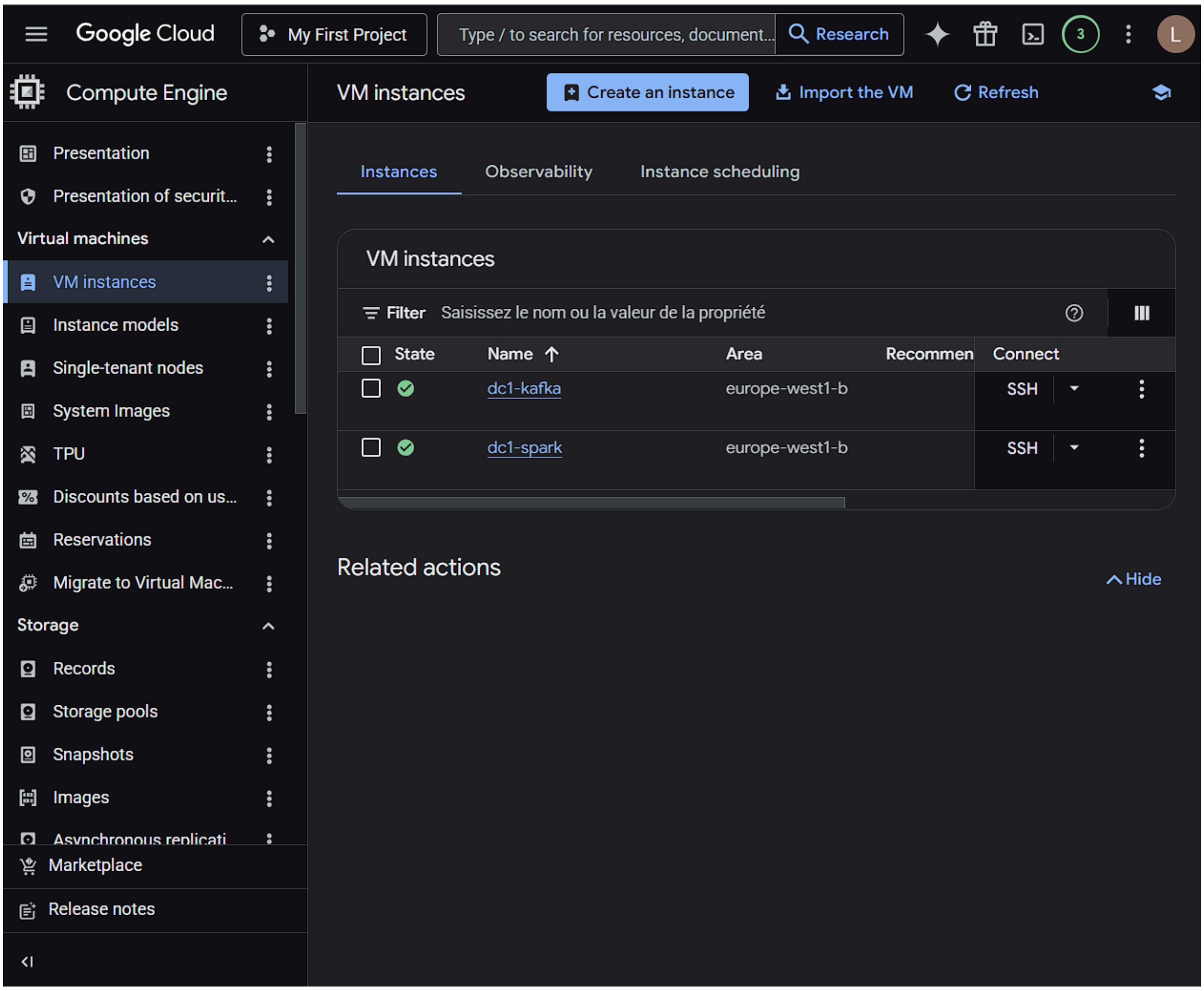

Google Cloud
My First Project
Type / to search for resources, document...
Research
Create an instance
Create a VM from...
Equivalent code
Machine configuration
e2-standard-2, europe-west1
OS and storage
Debian GNU/Linux 12 (bookworm)
Data protection
No backup
Networking
1 network interface
Observability
Install the Ops agent
Security
Advance
Machine configuration
Name *
dc1-spark
Region *
europe-west1 (Belgium)
The selection of a region is final.
Area *
All
Google will choose a zone for you to optimize your VM acquisition. The zone is final.
Generate machine type suggestions with Gemini
Prompt
Generate
Describe your workload and requirements (e.g., cost or performance). Generating machine type suggestions is free.
Low-cost web server
Low-volume database
AI/ML Inference
NEW: C4N series of machines now available for preview
Try C4N, the first instance of Google Cloud optimized for networking and block
Register
General use
Optimized for calculation
Optimized memory
Optimized for storage
GPU
Types of machines for common workloads to optimize costs and flexibility
Series
Description
vCPUs
Memory
C4
High and consistent performance
2 - 288
4 - 2,232 GB
Create
Cancel
Equivalent code

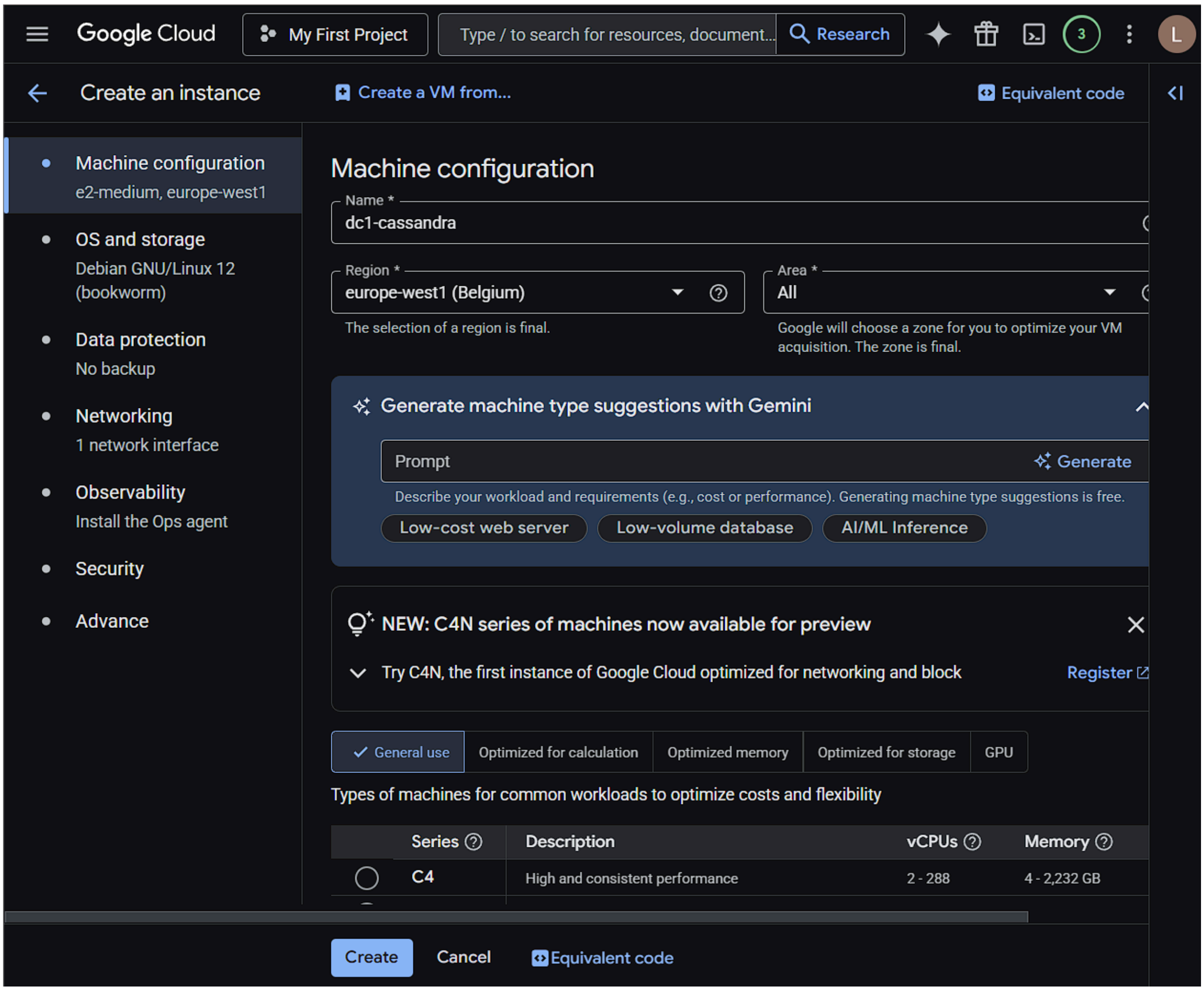

Google Cloud
My First Project
Type / to search for resources, document...
Research
Create an instance
Create a VM from...
Equivalent code
Machine configuration
e2-standard-2, europe-west1
OS and storage
Debian GNU/Linux 12 (bookworm)
Data protection
No backup
Networking
1 network interface
Observability
Install the Ops agent
Security
Advance
General use
Optimized for calculation
Optimized memory
Optimized for storage
GPU
Types of machines for common workloads to optimize costs and flexibility
Series | Description | vCPUs | Memory
C4 | High and consistent performance | 2 - 288 | 4 - 2,232 GB
C4A | High and consistent performance, Arm-based solution | 1 - 96 | 2 - 768 GB
C4D | High and consistent performance | 2 - 384 | 3 - 3072 GB
N4 | Flexible and economical | 2 - 80 | 4 - 640 GB
N4A | Flexibility and cost optimization based on Arm | 1 - 64 | 2 - 512 GB
N4D | Flexible and economical | 2 - 96 | 4 - 768 GB
C3 | High and consistent performance | 4 - 192 | 8 - 1536 GB
C3D | High and consistent performance | 4 - 360 | 8 - 2880 GB
E2 | Low-cost everyday computing | 0.25 - 32 | 1 - 128 GB
N2 | Balance between price and performance | 2 - 128 | 2 - 864 GB
N2D | Balance between price and performance | 2 - 224 | 2 - 896 GB
T2A | Horizontally scaling workloads | 1 - 48 | 4 - 192 GB
T2D | Horizontally scaling workloads | 1 - 60 | 4 - 240 GB
N1 | Balance between price and performance | 0.25 - 96 | 0.6 - 624 GB
Machine type
Choose a machine type with predefined amounts of vCPUs and memory suitable for most workloads. You can also create a custom machine for your specific workload needs. Learn more
Create
Cancel
Equivalent code

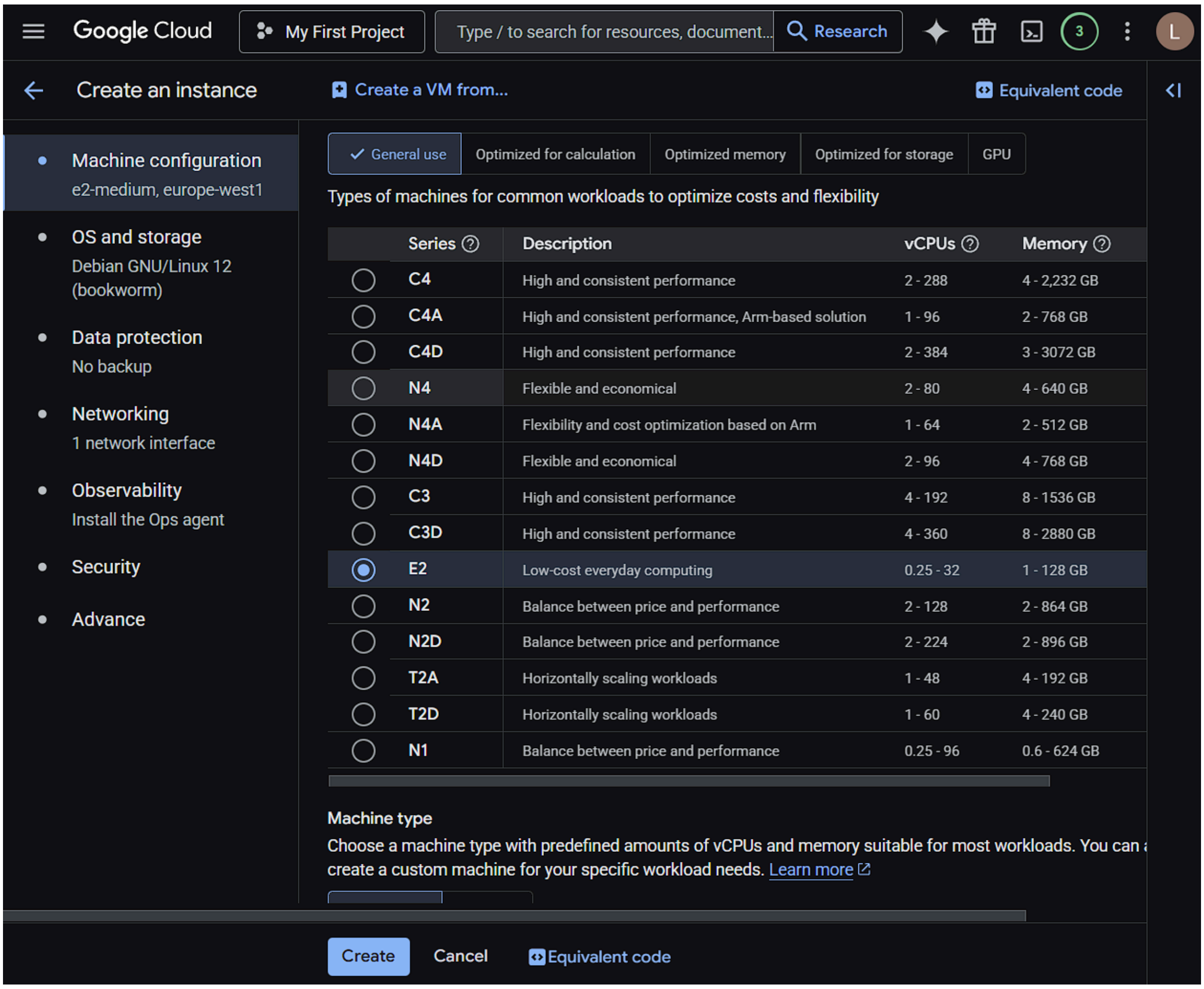

Google Cloud
My First Project
Type / to search for resources, document...
Research
3
L
Create an instance
Create a VM from...
Equivalent code
Machine configuration
e2-standard-2, europe-west1
OS and storage
Debian GNU/Linux 12 (bookworm)
Data protection
No backup
Networking
1 network interface
Observability
Install the Ops agent
Security
Advance
E2 Low-cost everyday computing 0.25 - 32 1 - 128 GB Inte
N2 Balance between price and performance 2 - 128 2 - 864 GB Inte
N2D Balance between price and performance 2 - 224 2 - 896 GB AM
T2A Horizontally scaling workloads 1 - 48 4 - 192 GB Am
T2D Horizontally scaling workloads 1 - 60 4 - 240 GB AM
N1 Balance between price and performance 0.25 - 96 0.6 - 624 GB Inte
Machine type
Choose a machine type with predefined amounts of vCPUs and memory suitable for most workloads. You can also create a custom machine for your specific workload needs. Learn more
Predefined
Custom
e2-standard-2 (2 vCPUs, 1 core, 8 GB of memory)
vCPU
2 (1 heart)
Memory
8 GB
Advanced configurations
Provisioning model
VM provisioning model
Standard
Determines the execution time of your VMs. This choice is final.
Advanced VM provisioning model settings
Create
Cancel
Equivalent code

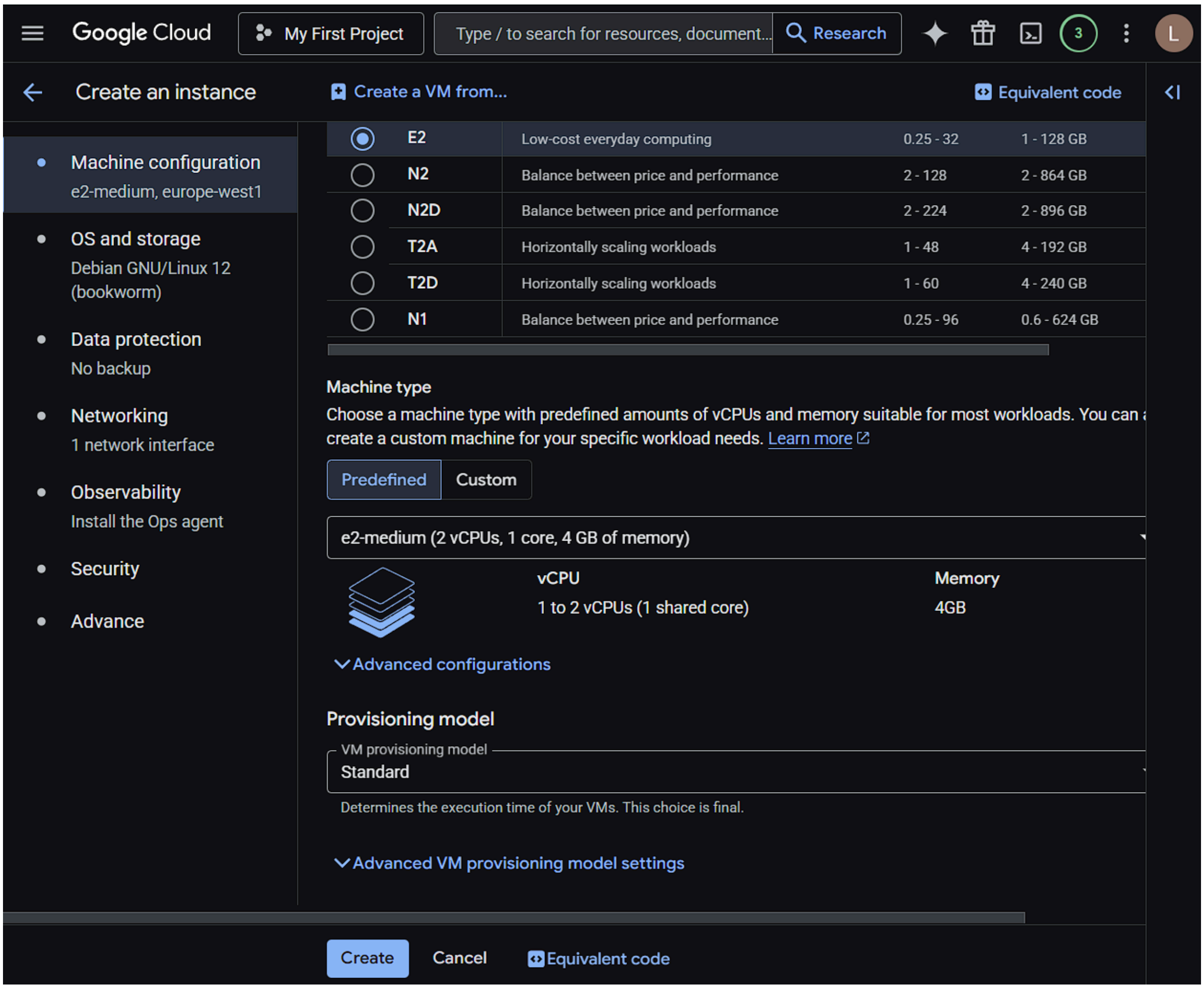

Boot disk
Select an image or snapshot to create a boot disk, or link an existing disk. Can't find what you're looking for? Explore hundreds of VM solutions in the Marketplace
public images
Custom images
Snapshots
Archival snapshots
Existing disks
Operating system
Debian
Version *
Ubuntu 26.04 LTS Minimal
x86/64, amd64 built on 20260428
Boot disk type *
Persistent disk with balancing
Compare the types of discs
Size (GB) *
10
Provision between 10 and 65536 GB
Show advanced configuration options
Select
Cancel

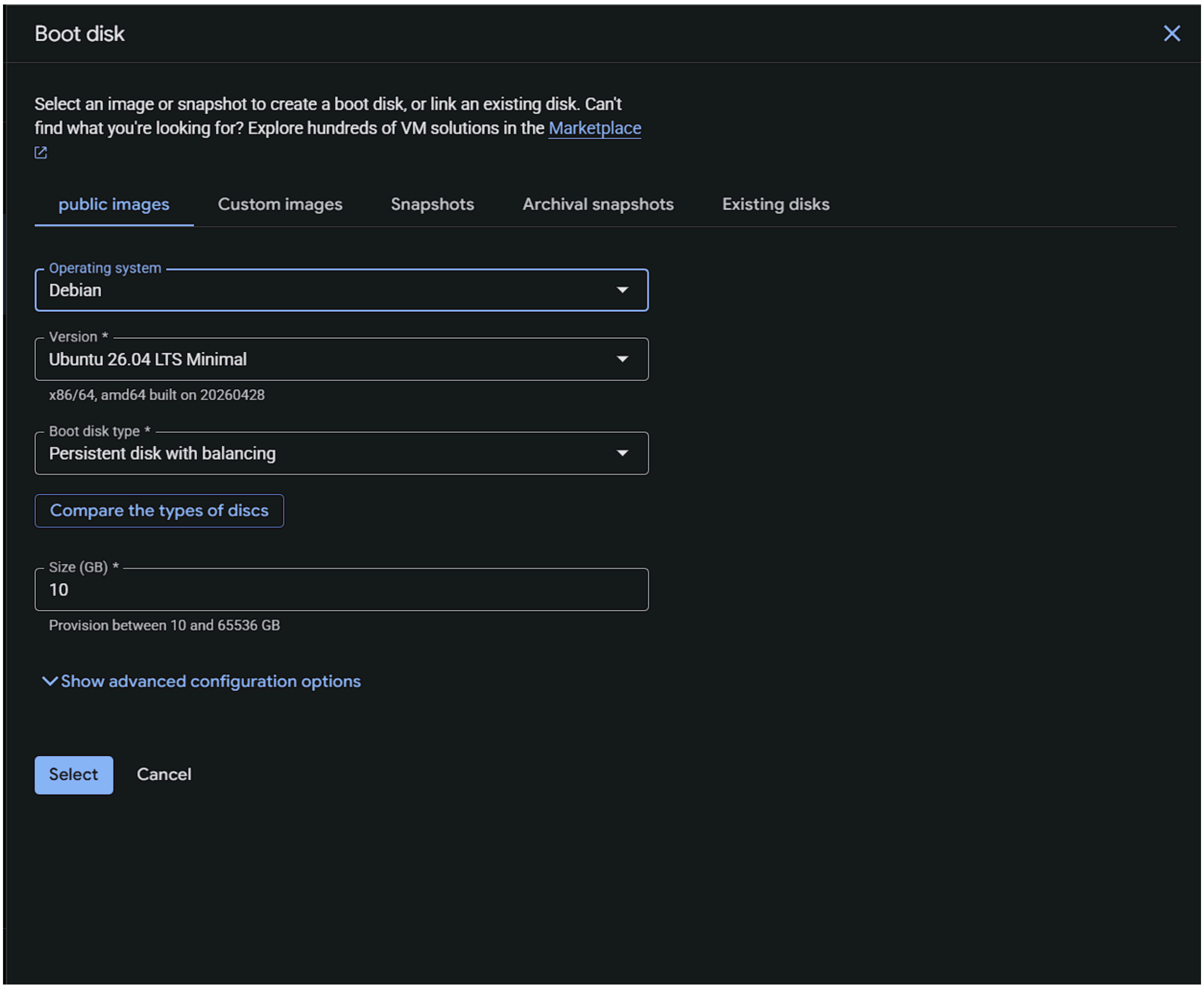

Google Cloud
My First Project
Type / to search for resources, document...
Research
Create an instance
Create a VM from...
Equivalent code
Machine configuration
e2-standard-2, europe-west1
OS and storage
Debian GNU/Linux 12 (bookworm)
Data protection
No backup
Networking
1 network interface
Observability
Security
Advance
Network
Firewall
Select firewall rules to allow specific network traffic from the internet. Firewall rules will only be applied when you create an instance.
Allow HTTP traffic
Allow HTTPS traffic
Allow load balancer status checks
Network tags
Network tags
http-server
https-server
Hostname
Hostname
Set a custom hostname for this instance or leave the default. This choice is final.
IP Transfer
Enable
Network performance configuration
Network bandwidth
Enable Tier_1 network performance per VM
Maximum outgoing network bandwidth: 4 Gbit/s
Monthly estimate
US$54.81
This equates to an hourly cost of approximately US$0.08
You pay for what you use: billing by the second, with no upfront fees.
Element
Monthly estimat
2 vCPU + 8 Go de mémoire
53,81 $U
Disque persistant avec équilibrage, d'une taille de 10 Go
1,00 $U
Programmation des instantanés
Coût variable
Total
54,81 $U
Compute Engine Pricing
Less
Create
Cancel
Equivalent code

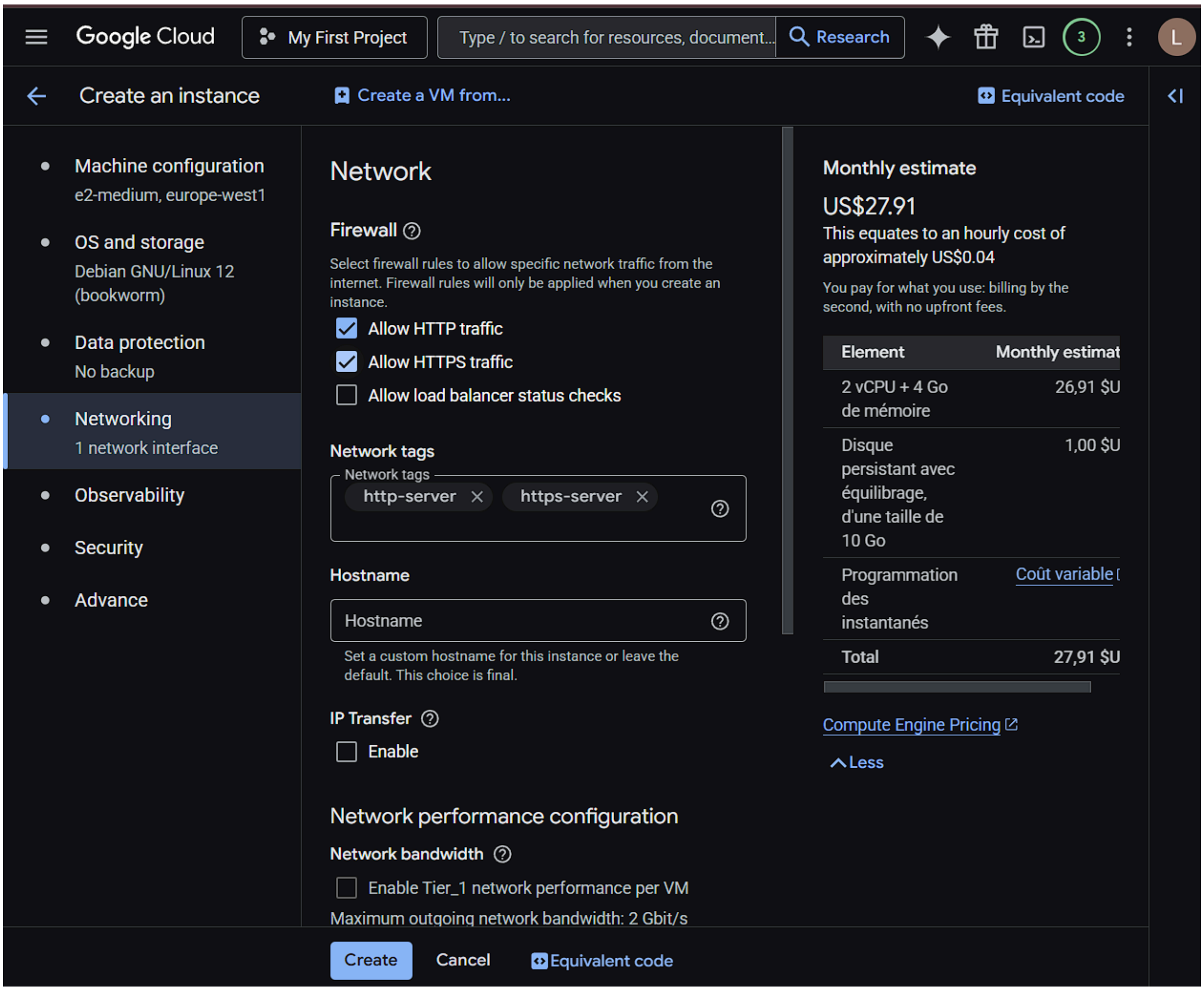

Google Cloud
My First Project
Type / to search for resources, document...
Research
Compute Engine
VM instances
Create an instance
Import the VM
Refresh
Presentation
Presentation of securit...
Virtual machines
VM instances
Instance models
Single-tenant nodes
System Images
TPU
Discounts based on us...
Reservations
Migrate to Virtual Mac...
Storage
Records
Storage pools
Snapshots
Images
Asynchronous replicati
Marketplace
Release notes
Instances
Observability
Instance scheduling
VM instances
Filter
Saisissez le nom ou la valeur de la propriété
State
Name
Area
Recommen
Connect
dc1-kafka
europe-west1-b
SSH
dc1-spark
europe-west1-b
SSH
Related actions
Hide

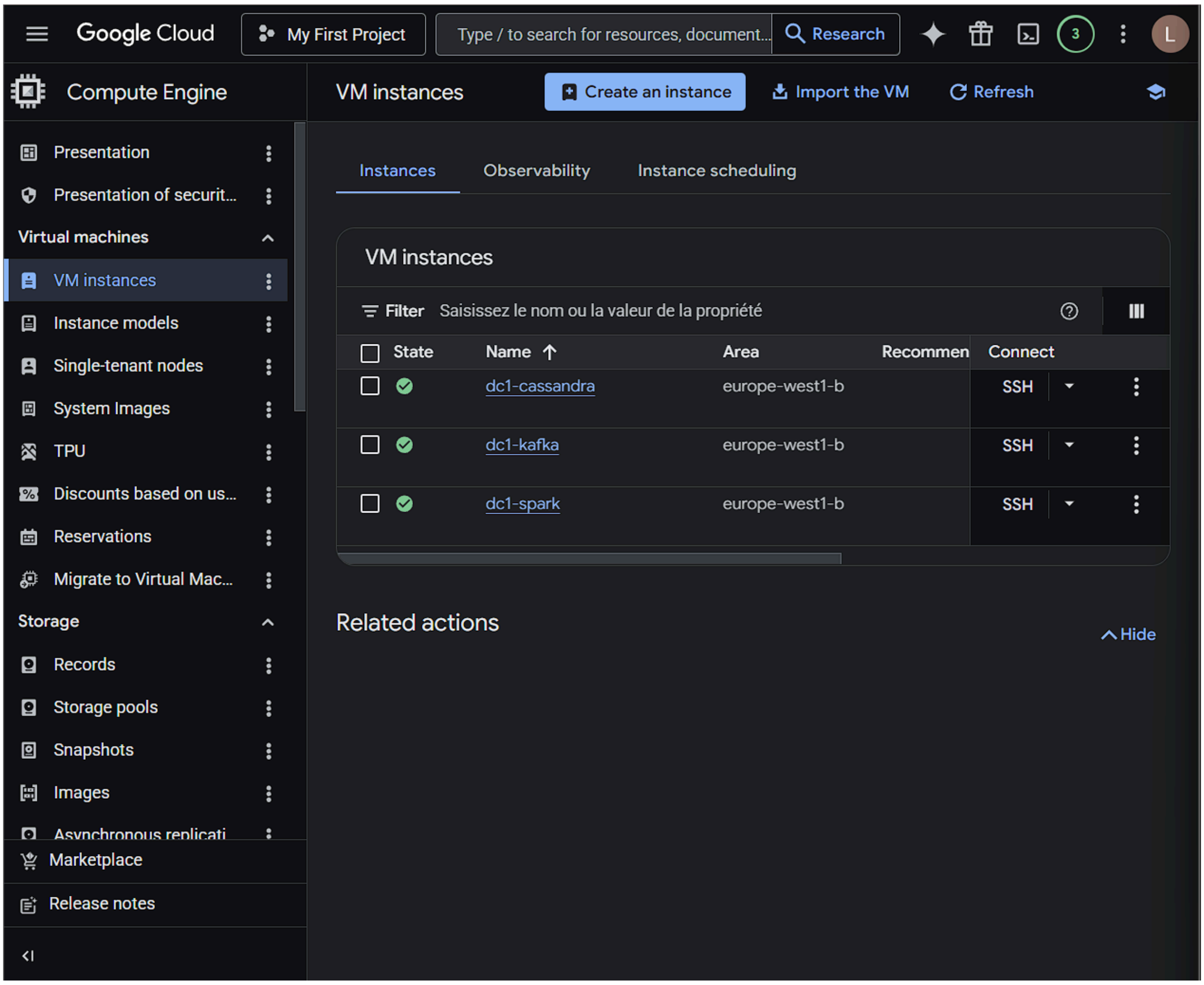

Google Cloud
My First Project
Type / to search for resources, document...
Research
Create an instance
Create a VM from...
Equivalent code
Machine configuration
e2-medium, europe-west1
OS and storage
Debian GNU/Linux 12 (bookworm)
Data protection
No backup
Networking
1 network interface
Observability
Install the Ops agent
Security
Advance
Machine configuration
Name *
dc1-cassandra
Region *
europe-west1 (Belgium)
The selection of a region is final.
Area *
All
Google will choose a zone for you to optimize your VM acquisition. The zone is final.
Generate machine type suggestions with Gemini
Prompt
Generate
Describe your workload and requirements (e.g., cost or performance). Generating machine type suggestions is free.
Low-cost web server
Low-volume database
AI/ML Inference
NEW: C4N series of machines now available for preview
Try C4N, the first instance of Google Cloud optimized for networking and block
Register
General use
Optimized for calculation
Optimized memory
Optimized for storage
GPU
Types of machines for common workloads to optimize costs and flexibility
Series
Description
vCPUs
Memory
C4
High and consistent performance
2 - 288
4 - 2,232 GB
Create
Cancel
Equivalent code


Google Cloud
My First Project
Type / to search for resources, document...
Research
Create an instance
Create a VM from...
Equivalent code
Machine configuration
e2-medium, europe-west1
OS and storage
Debian GNU/Linux 12 (bookworm)
Data protection
No backup
Networking
1 network interface
Observability
Install the Ops agent
Security
Advance
General use
Optimized for calculation
Optimized memory
Optimized for storage
GPU
Types of machines for common workloads to optimize costs and flexibility
Series | Description | vCPUs | Memory
C4 | High and consistent performance | 2 - 288 | 4 - 2,232 GB
C4A | High and consistent performance, Arm-based solution | 1 - 96 | 2 - 768 GB
C4D | High and consistent performance | 2 - 384 | 3 - 3072 GB
N4 | Flexible and economical | 2 - 80 | 4 - 640 GB
N4A | Flexibility and cost optimization based on Arm | 1 - 64 | 2 - 512 GB
N4D | Flexible and economical | 2 - 96 | 4 - 768 GB
C3 | High and consistent performance | 4 - 192 | 8 - 1536 GB
C3D | High and consistent performance | 4 - 360 | 8 - 2880 GB
E2 | Low-cost everyday computing | 0.25 - 32 | 1 - 128 GB
N2 | Balance between price and performance | 2 - 128 | 2 - 864 GB
N2D | Balance between price and performance | 2 - 224 | 2 - 896 GB
T2A | Horizontally scaling workloads | 1 - 48 | 4 - 192 GB
T2D | Horizontally scaling workloads | 1 - 60 | 4 - 240 GB
N1 | Balance between price and performance | 0.25 - 96 | 0.6 - 624 GB
Machine type
Choose a machine type with predefined amounts of vCPUs and memory suitable for most workloads. You can a
create a custom machine for your specific workload needs. Learn more
Create
Cancel
Equivalent code


Google Cloud
My First Project
Type / to search for resources, document...
Research
Create an instance
Create a VM from...
Equivalent code
Machine configuration
e2-medium, europe-west1
OS and storage
Debian GNU/Linux 12 (bookworm)
Data protection
No backup
Networking
1 network interface
Observability
Install the Ops agent
Security
Advance
E2 Low-cost everyday computing 0.25 - 32 1 - 128 GB
N2 Balance between price and performance 2 - 128 2 - 864 GB
N2D Balance between price and performance 2 - 224 2 - 896 GB
T2A Horizontally scaling workloads 1 - 48 4 - 192 GB
T2D Horizontally scaling workloads 1 - 60 4 - 240 GB
N1 Balance between price and performance 0.25 - 96 0.6 - 624 GB
Machine type
Choose a machine type with predefined amounts of vCPUs and memory suitable for most workloads. You can
create a custom machine for your specific workload needs. Learn more
Predefined
Custom
e2-medium (2 vCPUs, 1 core, 4 GB of memory)
vCPU
1 to 2 vCPUs (1 shared core)
Memory
4GB
Advanced configurations
Provisioning model
VM provisioning model
Standard
Determines the execution time of your VMs. This choice is final.
Advanced VM provisioning model settings
Create
Cancel
Equivalent code


Boot disk
Select an image or snapshot to create a boot disk, or link an existing disk. Can't find what you're looking for? Explore hundreds of VM solutions in the Marketplace
public images
Custom images
Snapshots
Archival snapshots
Existing disks
Operating system
Debian
Version *
Ubuntu 26.04 LTS Minimal
x86/64, amd64 built on 20260428
Boot disk type *
Persistent disk with balancing
Compare the types of discs
Size (GB) *
10
Provision between 10 and 65536 GB
Show advanced configuration options
Select
Cancel


Google Cloud
My First Project
Type / to search for resources, document...
Research
Create an instance
Create a VM from...
Equivalent code
Machine configuration
e2-medium, europe-west1
OS and storage
Debian GNU/Linux 12 (bookworm)
Data protection
No backup
Networking
1 network interface
Observability
Security
Advance
Network
Firewall
Select firewall rules to allow specific network traffic from the internet. Firewall rules will only be applied when you create an instance.
Allow HTTP traffic
Allow HTTPS traffic
Allow load balancer status checks
Network tags
Network tags
http-server
https-server
Hostname
Hostname
Set a custom hostname for this instance or leave the default. This choice is final.
IP Transfer
Enable
Network performance configuration
Network bandwidth
Enable Tier_1 network performance per VM
Maximum outgoing network bandwidth: 2 Gbit/s
Monthly estimate
US$27.91
This equates to an hourly cost of approximately US$0.04
You pay for what you use: billing by the second, with no upfront fees.
Element
Monthly estimat
2 vCPU + 4 Go de mémoire
26,91 $U
Disque persistant avec équilibrage, d'une taille de 10 Go
1,00 $U
Programmation des instantanés
Coût variable
Total
27,91 $U
Compute Engine Pricing
Less
Create
Cancel
Equivalent code


Google Cloud
My First Project
Type / to search for resources, document...
Research
Compute Engine
VM instances
Create an instance
Import the VM
Refresh
Presentation
Presentation of securit...
Virtual machines
VM instances
Instance models
Single-tenant nodes
System Images
TPU
Discounts based on us...
Reservations
Migrate to Virtual Mac...
Storage
Records
Storage pools
Snapshots
Images
Marketplace
Release notes
Instances
Observability
Instance scheduling
VM instances
Filter
Saisissez le nom ou la valeur de la propriété
State
Name
Area
Recommen
Connect
dc1-cassandra
europe-west1-b
SSH
dc1-kafka
europe-west1-b
SSH
dc1-spark
europe-west1-b
SSH
Related actions
Hide

Cloud Shell Editor

(project-c15907ce-0bab-4a45-b70)

```
Processing triggers for libc-bin (2.43-2ubuntu2) ...
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-kafka --zone=europe-west1-b --com
mand="sudo apt-get update && sudo apt-get install -y docker.io docker-compose && sudo usermod -aG docker larissa_mbonazolusakuen
o"
Warning: Permanently added 'compute.7766001525143635847' (ED25519) to the list of known hosts.
Hit:1 http://europe-west1.gce.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://europe-west1.gce.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://europe-west1.gce.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:5 https://packages.cloud.google.com/apt google-cloud-ops-agent-jammy-2 InRelease
Reading package lists...
Reading package lists...
Building dependency tree...
Reading state information...
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base git git-man iptables less
  libcurl3-gnutls liberror-perl libip6tc2 libjansson4 libnetfilter-conntrack3
  libnfnetlink0 libnftables1 libnftnl11 netcat netcat-openbsd pigz
  python3-docker python3-dockerpty python3-docopt python3-dotenv
  python3-texttable python3-websocket runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx
  docker-compose-v2 docker-doc rinse zfs-fuse | zfsutils git-daemon-run
  | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb git-cvs
  git-mediawiki git-svn firewalld nftables
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker-compose docker.io
  git git-man iptables less libcurl3-gnutls liberror-perl libip6tc2
  libjansson4 libnetfilter-conntrack3 libnfnetlink0 libnftables1 libnftnl11
  netcat netcat-openbsd pigz python3-docker python3-dockerpty python3-docopt
  python3-dotenv python3-texttable python3-websocket runc ubuntu-fan
0 upgraded, 29 newly installed, 0 to remove and 9 not upgraded.
Need to get 79.9 MB of archives.
After this operation, 307 MB of additional disk space will be used.
Get:1 http://europe-west1.gce.archive.ubuntu.com/ubuntu jammy/universe amd64 pigz amd64 2.6-1 [63.6 kB]
Get:2 http://europe-west1.gce.archive.ubuntu.com/ubuntu jammy-updates/main amd64 less amd64 590-1ubuntu0.22.04.3 [142 kB]
Get:3 http://europe-west1.gce.archive.ubuntu.com/ubuntu jammy/main amd64 netcat-openbsd amd64 1.218-4ubuntu1 [39.4 kB]
Get:4 http://europe-west1.gce.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libip6tc2 amd64 1.8.7-1ubuntu5.2 [20.3 kB]
Get:5 http://europe-west1.gce.archive.ubuntu.com/ubuntu jammy/main amd64 libnfnetlink0 amd64 1.0.1-3build3 [14.6 kB]
Get:6 http://europe-west1.gce.archive.ubuntu.com/ubuntu jammy/main amd64 libnetfilter-conntrack3 amd64 1.0.9-1 [45.3 kB]
Get:7 http://europe-west1.gce.archive.ubuntu.com/ubuntu jammy/main amd64 libnftnl11 amd64 1.2.1-1build1 [65.5 kB]
Get:8 http://europe-west1.gce.archive.ubuntu.com/ubuntu jammy-updates/main amd64 iptables amd64 1.8.7-1ubuntu5.2 [455 kB]
Get:9 http://europe-west1.gce.archive.ubuntu.com/ubuntu jammy/main amd64 libjansson4 amd64 2.13.1-1.1build3 [32.4 kB]
Get:10 http://europe-west1.gce.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libnftables1 amd64 1.0.2-1ubuntu3.1 [332 kB]
Get:11 http://europe-west1.gce.archive.ubuntu.com/ubuntu jammy/main amd64 bridge-utils amd64 1.7-1ubuntu3 [34.4 kB]
Get:12 http://europe-west1.gce.archive.ubuntu.com/ubuntu jammy-updates/main amd64 runc amd64 1.3.4-0ubuntu1~22.04.1 [9569 kB]
Get:13 http://europe-west1.gce.archive.ubuntu.com/ubuntu jammy-updates/main amd64 containerd amd64 2.2.1-0ubuntu1~22.04.1 [28.2
```

Cloud Shell Editor

(project-c15907ce-0bab-4a45-b70)

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-spark --zone=europe-west1-b --com
mand="sudo apt-get update && sudo apt-get install -y docker.io docker-compose && sudo usermod -aG docker larissa_mbonazolusakuen
o"
Warning: Permanently added 'compute.8259961663702434865' (ED25519) to the list of known hosts.
Get:1 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:2 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute-updates InRelease [136 kB]
Get:3 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security InRelease [136 kB]
Get:5 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute/main amd64v3 Packages [1483 kB]
Get:6 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute/main Translation-en [524 kB]
Get:7 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [18.5 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [9748 B]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64v3 Packages [27.2 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [8072 B]
Get:12 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:13 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute/restricted amd64v3 Packages [152 kB]
Get:14 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute/restricted Translation-en [25.8 kB]
Get:15 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute/multiverse amd64v3 Packages [291 kB]
Get:16 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute/multiverse Translation-en [127 kB]
Get:17 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages [54.0 kB]
Get:18 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [15.4 kB]
Get:19 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 Packages [27.2 kB]
Get:20 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [8072 B]
Get:21 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64v3 Packages [29.4 kB]
Get:22 http://europe-west1.gce.archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [6588 B]
Fetched 25.9 MB in 3s (7459 kB/s)
Reading package lists...
Reading package lists...
Building dependency tree...
Reading state information...
Solving dependencies...
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base git git-man iptables less
  liberror-perl libgdbm-compat4t64 libgdbm6t64 libip4tc2 libip6tc2 libjansson4
  libnetfilter-conntrack3 libnfnetlink0 libnftables1 libnftnl11 libperl5.40
  nftables perl perl-modules-5.40 pigz runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools btrfs-progs cgroupfs-mount | cgroup-lite debootstrap
  docker-buildx docker-doc rinse rootlesskit zfs-fuse | zfsutils git-doc
  git-email git-gui gitk gitweb git-cvs git-svn firewalld gdbm-l10n perl-doc
  libterm-readline-gnu-perl | libterm-readline-perl-perl make
  libtap-harness-archive-perl
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker-compose-v2
```

```
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-cassandra --zone=europe-west1-b -
-command="sudo docker run hello-world"
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pulling fs layer
d5e71e642bf5: Download complete
4f55086f7dd0: Download complete
4f55086f7dd0: Pull complete
Digest: sha256:f9078146db2e05e794366b1bfe584a14ea6317f44027d10ef7dad65279026885
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$
```

```
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-kafka --zone=europe-west1-b --com
mand="sudo docker run hello-world"
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pulling fs layer
d5e71e642bf5: Download complete
4f55086f7dd0: Download complete
4f55086f7dd0: Pull complete
Digest: sha256:f9078146db2e05e794366b1bfe584a14ea6317f44027d10ef7dad65279026885
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

```
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-spark --zone=europe-west1-b --com
mand="sudo docker run hello-world"
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pulling fs layer
d5e71e642bf5: Download complete
4f55086f7dd0: Download complete
4f55086f7dd0: Pull complete
Digest: sha256:f9078146db2e05e794366b1bfe584a14ea6317f44027d10ef7dad65279026885
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

```
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute instances list
NAME: dc1-cassandra
ZONE: europe-west1-b
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.4
EXTERNAL_IP: 34.22.143.37
STATUS: RUNNING

NAME: dc1-kafka
ZONE: europe-west1-b
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.2
EXTERNAL_IP: 35.205.160.140
STATUS: RUNNING

NAME: dc1-spark
ZONE: europe-west1-b
MACHINE_TYPE: e2-standard-2
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.3
EXTERNAL_IP: 34.140.166.171
STATUS: RUNNING
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$
```

```
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute firewall-rules create allow-eurovision \
> --allow tcp:9092,tcp:9042,tcp:7077,tcp:8080,tcp:2181,tcp:4040 \
> --source-ranges 0.0.0.0/0 \
> --description "Eurovision pipeline ports"
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/project-c15907ce-0bab-4a45-b70/global/firew
alls/allow-eurovision].
Creating firewall...done.
NAME: allow-eurovision
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:9092,tcp:9042,tcp:7077,tcp:8080,tcp:2181,tcp:4040
DENY:
DISABLED: False
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$
```

Cloud Shell Editor

(project-c15907ce-0bab-4a45-b70)

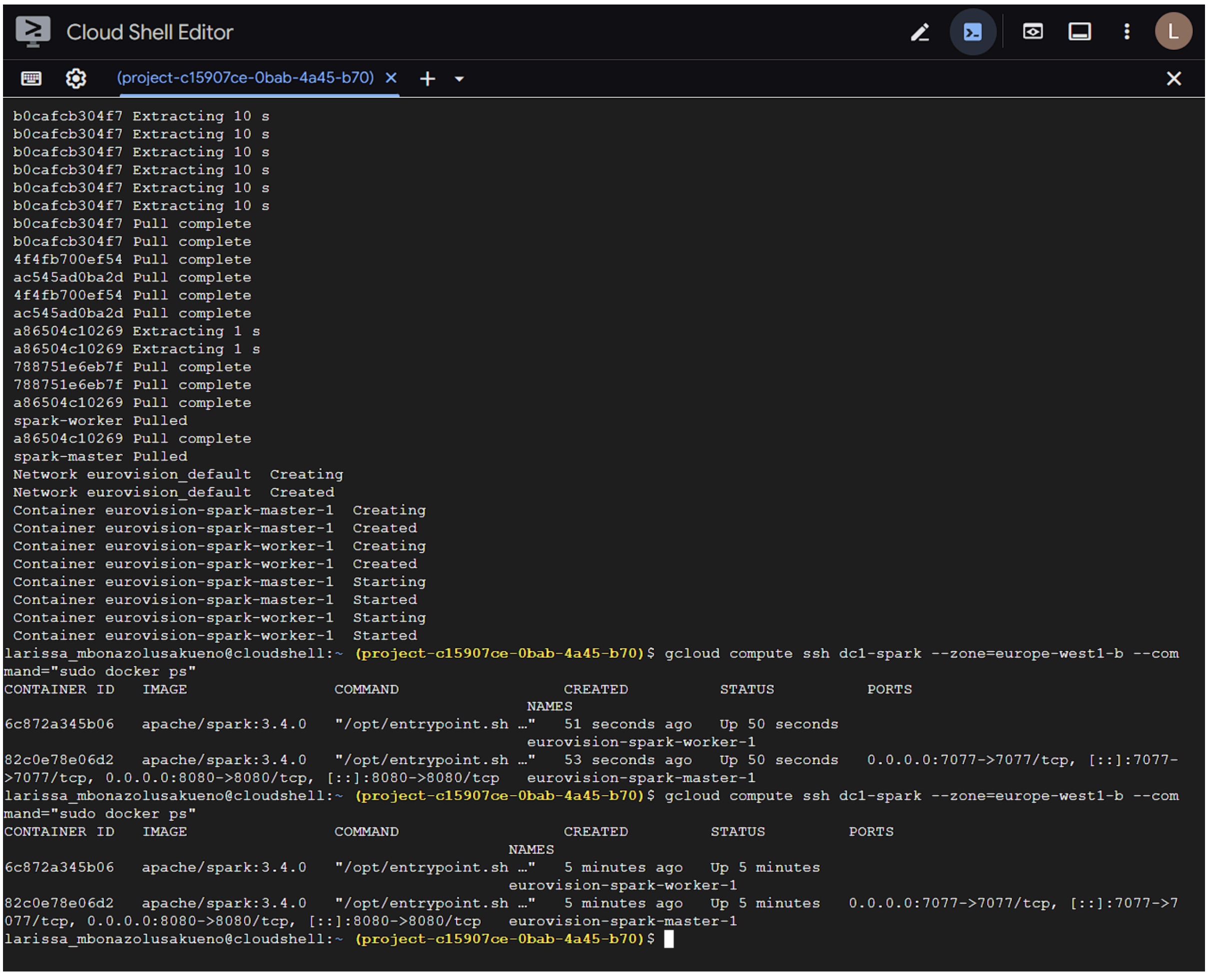
```
> --source-ranges 0.0.0.0/0 \
> --description "Eurovision pipeline ports"
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/project-c15907ce-0bab-4a45-b70/global/firew
alls/allow-eurovision].
Creating firewall...done.
NAME: allow-eurovision
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:9092,tcp:9042,tcp:7077,tcp:8080,tcp:2181,tcp:4040
DENY:
DISABLED: False
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-kafka --zone=europe-west1-b --com
mand="mkdir -p ~/eurovision"
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-kafka --zone=europe-west1-b --com
mand="cat > ~/eurovision/docker-compose.yml << 'EOF'
> version: '3'
> services:
> zookeeper:
> image: confluentinc/cp-zookeeper:7.4.0
> ^C
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-kafka --zone=europe-west1-b --com
mand="printf 'version: \"3\"\nservices:\n  zookeeper:\n    image: confluentinc/cp-zookeeper:7.4.0\n    environment:\n      ZOOKE
EPER_CLIENT_PORT: 2181\n    restart: always\n  kafka:\n    image: confluentinc/cp-kafka:7.4.0\n    depends_on: [zookeeper]\n
ports:\n      - \"9092:9092\"\n    environment:\n      KAFKA_ZOOKEEPER_CONNECT: zookeeper:2181\n      KAFKA_ADVERTISED_LISTENERS
: PLAINTEXT://35.205.160.140:9092\n      KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR: 1\n    restart: always\n' > ~/eurovision/docker
-compose.yml"
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-kafka --zone=europe-west1-b --com
mand="cat ~/eurovision/docker-compose.yml"
version: "3"
services:
  zookeeper:
    image: confluentinc/cp-zookeeper:7.4.0
    environment:
      ZOOKEEPER_CLIENT_PORT: 2181
    restart: always
  kafka:
    image: confluentinc/cp-kafka:7.4.0
    depends_on: [zookeeper]
    ports:
      - "9092:9092"
    environment:
      KAFKA_ZOOKEEPER_CONNECT: zookeeper:2181
      KAFKA_ADVERTISED_LISTENERS: PLAINTEXT://35.205.160.140:9092
      KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR: 1
    restart: always
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$
```

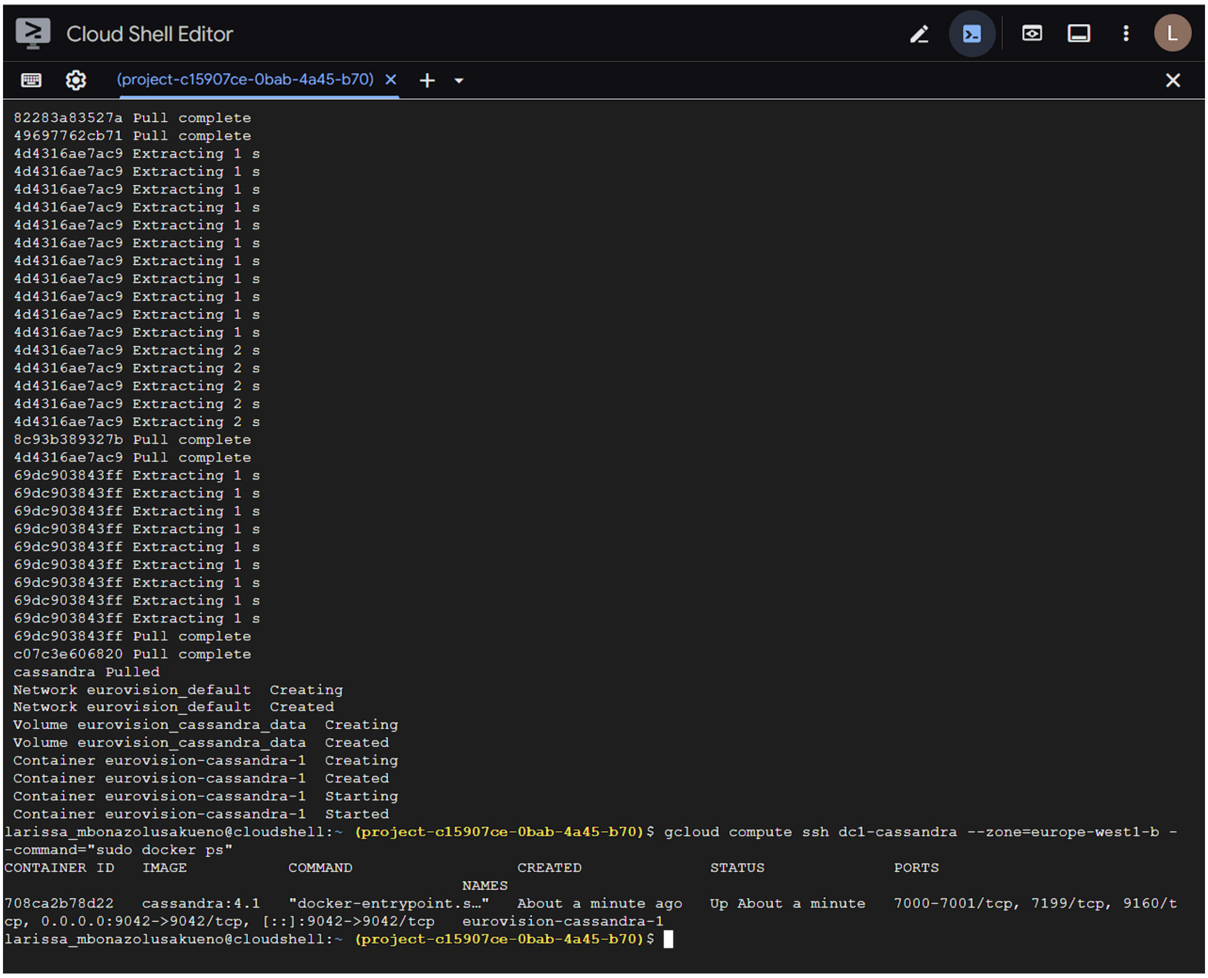
Cloud Shell Editor

(project-c15907ce-0bab-4a45-b70)

```
EPER_CLIENT_PORT: 2181\n    restart: always\n  kafka:\n    image: confluentinc/cp-kafka:7.4.0\n    depends_on: [zookeeper]\n
ports:\n      - \"9092:9092\"\n    environment:\n      KAFKA_ZOOKEEPER_CONNECT: zookeeper:2181\n      KAFKA_ADVERTISED_LISTENERS
: PLAINTEXT://35.205.160.140:9092\n      KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR: 1\n    restart: always\n' > ~/eurovision/docker
-compose.yml"
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-kafka --zone=europe-west1-b --com
mand="cat ~/eurovision/docker-compose.yml"
version: "3"
services:
  zookeeper:
    image: confluentinc/cp-zookeeper:7.4.0
    environment:
      ZOOKEEPER_CLIENT_PORT: 2181
    restart: always
  kafka:
    image: confluentinc/cp-kafka:7.4.0
    depends_on: [zookeeper]
    ports:
      - "9092:9092"
    environment:
      KAFKA_ZOOKEEPER_CONNECT: zookeeper:2181
      KAFKA_ADVERTISED_LISTENERS: PLAINTEXT://35.205.160.140:9092
      KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR: 1
    restart: always
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-kafka --zone=europe-west1-b --com
mand="cd ~/eurovision && sudo docker-compose up -d"
Creating network "eurovision_default" with the default driver
Pulling zookeeper (confluentinc/cp-zookeeper:7.4.0)...
7.4.0: Pulling from confluentinc/cp-zookeeper
Digest: sha256:90631f224b4397ecfb4e824e43d93e0e42656841c6d55ef635c49df3975260ae
Status: Downloaded newer image for confluentinc/cp-zookeeper:7.4.0
Pulling kafka (confluentinc/cp-kafka:7.4.0)...
7.4.0: Pulling from confluentinc/cp-kafka
Digest: sha256:187dac6627e7906c350f5f8c982f80ce735ff1a0e571a20de6000a309a12ce63
Status: Downloaded newer image for confluentinc/cp-kafka:7.4.0
Creating eurovision_zookeeper_1 ...
Creating eurovision_zookeeper_1 ... done
Creating eurovision_kafka_1     ...
Creating eurovision_kafka_1     ... done
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-kafka --zone=europe-west1-b --com
mand="sudo docker ps"
CONTAINER ID   IMAGE                             COMMAND                  CREATED         STATUS         PORTS
                   NAMES
d0b8dcf971ba   confluentinc/cp-kafka:7.4.0       "/etc/confluent/dock…"   2 minutes ago   Up 2 minutes   0.0.0.0:9092->9092/tcp,
 [::]:9092->9092/tcp   eurovision_kafka_1
b12077222214   confluentinc/cp-zookeeper:7.4.0   "/etc/confluent/dock…"   3 minutes ago   Up 2 minutes   2181/tcp, 2888/tcp, 388
8/tcp                  eurovision_zookeeper_1
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$
```

```
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-spark --zone=europe-west1-b --com
mand="mkdir -p ~/eurovision"
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-spark --zone=europe-west1-b --com
mand="printf 'version: \"3\"\nservices:\n  spark-master:\n    image: bitnami/spark:3.4\n    ports:\n      - \"8080:8080\"\n
 - \"7077:7077\"\n    environment:\n      - SPARK_MODE=master\n    restart: always\n  spark-worker:\n    image: bitnami/spark:3.
4\n    depends_on: [spark-master]\n    environment:\n      - SPARK_MODE=worker\n      - SPARK_MASTER_URL=spark://spark-master:70
77\n    restart: always\n' > ~/eurovision/docker-compose.yml"
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-spark --zone=europe-west1-b --com
mand="cat ~/eurovision/docker-compose.yml"
version: "3"
services:
  spark-master:
    image: bitnami/spark:3.4
    ports:
      - "8080:8080"
      - "7077:7077"
    environment:
      - SPARK_MODE=master
    restart: always
  spark-worker:
    image: bitnami/spark:3.4
    depends_on: [spark-master]
    environment:
      - SPARK_MODE=worker
      - SPARK_MASTER_URL=spark://spark-master:7077
    restart: always
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-spark --zone=europe-west1-b --com
mand="cd ~/eurovision && sudo docker-compose up -d"
sudo: 'docker-compose': command not found
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$
```


Cloud Shell Editor
(project-c15907ce-0bab-4a45-b70)
b0cafcb304f7 Extracting 10 s
b0cafcb304f7 Extracting 10 s
b0cafcb304f7 Extracting 10 s
b0cafcb304f7 Extracting 10 s
b0cafcb304f7 Extracting 10 s
b0cafcb304f7 Extracting 10 s
b0cafcb304f7 Pull complete
b0cafcb304f7 Pull complete
4f4fb700ef54 Pull complete
ac545ad0ba2d Pull complete
4f4fb700ef54 Pull complete
ac545ad0ba2d Pull complete
a86504c10269 Extracting 1 s
a86504c10269 Extracting 1 s
788751e6eb7f Pull complete
788751e6eb7f Pull complete
a86504c10269 Pull complete
spark-worker Pulled
a86504c10269 Pull complete
spark-master Pulled
Network eurovision_default Creating
Network eurovision_default Created
Container eurovision-spark-master-1 Creating
Container eurovision-spark-master-1 Created
Container eurovision-spark-worker-1 Creating
Container eurovision-spark-worker-1 Created
Container eurovision-spark-master-1 Starting
Container eurovision-spark-master-1 Started
Container eurovision-spark-worker-1 Starting
Container eurovision-spark-worker-1 Started
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-spark --zone=europe-west1-b --com
mand="sudo docker ps"
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
NAMES
6c872a345b06 apache/spark:3.4.0 "/opt/entrypoint.sh …" 51 seconds ago Up 50 seconds
eurovision-spark-worker-1
82c0e78e06d2 apache/spark:3.4.0 "/opt/entrypoint.sh …" 53 seconds ago Up 50 seconds 0.0.0.0:7077->7077/tcp, [::]:7077-
>7077/tcp, 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp eurovision-spark-master-1
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-spark --zone=europe-west1-b --com
mand="sudo docker ps"
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
NAMES
6c872a345b06 apache/spark:3.4.0 "/opt/entrypoint.sh …" 5 minutes ago Up 5 minutes
eurovision-spark-worker-1
82c0e78e06d2 apache/spark:3.4.0 "/opt/entrypoint.sh …" 5 minutes ago Up 5 minutes 0.0.0.0:7077->7077/tcp, [::]:7077->7
077/tcp, 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp eurovision-spark-master-1
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$



```
Cloud Shell Editor
(project-c15907ce-0bab-4a45-b70)
82283a83527a Pull complete
49697762cb71 Pull complete
4d4316ae7ac9 Extracting 1 s
4d4316ae7ac9 Extracting 1 s
4d4316ae7ac9 Extracting 1 s
4d4316ae7ac9 Extracting 1 s
4d4316ae7ac9 Extracting 1 s
4d4316ae7ac9 Extracting 1 s
4d4316ae7ac9 Extracting 1 s
4d4316ae7ac9 Extracting 1 s
4d4316ae7ac9 Extracting 1 s
4d4316ae7ac9 Extracting 1 s
4d4316ae7ac9 Extracting 1 s
4d4316ae7ac9 Extracting 2 s
4d4316ae7ac9 Extracting 2 s
4d4316ae7ac9 Extracting 2 s
4d4316ae7ac9 Extracting 2 s
4d4316ae7ac9 Extracting 2 s
8c93b389327b Pull complete
4d4316ae7ac9 Pull complete
69dc903843ff Extracting 1 s
69dc903843ff Extracting 1 s
69dc903843ff Extracting 1 s
69dc903843ff Extracting 1 s
69dc903843ff Extracting 1 s
69dc903843ff Extracting 1 s
69dc903843ff Extracting 1 s
69dc903843ff Extracting 1 s
69dc903843ff Extracting 1 s
69dc903843ff Pull complete
c07c3e606820 Pull complete
cassandra Pulled
Network eurovision_default  Creating
Network eurovision_default  Created
Volume eurovision_cassandra_data  Creating
Volume eurovision_cassandra_data  Created
Container eurovision-cassandra-1  Creating
Container eurovision-cassandra-1  Created
Container eurovision-cassandra-1  Starting
Container eurovision-cassandra-1  Started
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-cassandra --zone=europe-west1-b -
-command="sudo docker ps"
CONTAINER ID   IMAGE           COMMAND                  CREATED              STATUS              PORTS
                                                   NAMES
708ca2b78d22   cassandra:4.1   "docker-entrypoint.s…"   About a minute ago   Up About a minute   7000-7001/tcp, 7199/tcp, 9160/t
cp, 0.0.0.0:9042->9042/tcp, [::]:9042->9042/tcp   eurovision-cassandra-1
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$
```

```
mand="echo 'RUN pip install kafka-python' >> ~/eurovision/producer/Dockerfile"
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-kafka --zone=europe-west1-b --com
mand="echo 'COPY producer.py .' >> ~/eurovision/producer/Dockerfile"
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-kafka --zone=europe-west1-b --com
mand="echo 'CMD [\"python\", \"producer.py\"]' >> ~/eurovision/producer/Dockerfile"
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-kafka --zone=europe-west1-b --com
mand="cd ~/eurovision/producer && sudo docker build -t eurovision-producer ."
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.584kB
Step 1/5 : FROM python:3.10-slim
3.10-slim: Pulling from library/python
6f7b1a1af1d2: Pulling fs layer
3531af2bc2a9: Pulling fs layer
b05032fa5b62: Pulling fs layer
b4e72df0ba0e: Pulling fs layer
706754b49547: Download complete
d8377d858729: Download complete
6f7b1a1af1d2: Download complete
b4e72df0ba0e: Download complete
b05032fa5b62: Download complete
3531af2bc2a9: Download complete
3531af2bc2a9: Pull complete
b4e72df0ba0e: Pull complete
6f7b1a1af1d2: Pull complete
b05032fa5b62: Pull complete
Digest: sha256:cdbf8193cee2e31639ea8ea85ffdd8fa5cce98ee9abfde96ea5f329490048831
Status: Downloaded newer image for python:3.10-slim
 ---> cdbf8193cee2
Step 2/5 : WORKDIR /app
 ---> Running in 472e58e70b5b
 ---> Removed intermediate container 472e58e70b5b
 ---> f024f3f6c55b
Step 3/5 : RUN pip install kafka-python
 ---> Running in 760b21982dff
Collecting kafka-python
  Downloading kafka_python-2.3.1-py2.py3-none-any.whl (326 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 326.1/326.1 kB 14.7 MB/s eta 0:00:00
Installing collected packages: kafka-python
Successfully installed kafka-python-2.3.1
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manag
er. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv

[notice] A new release of pip is available: 23.0.1 -> 26.1.1
[notice] To update, run: pip install --upgrade pip
```

```
Sending build context to Docker daemon  3.584kB
Step 1/5 : FROM python:3.10-slim
3.10-slim: Pulling from library/python
6f7b1a1af1d2: Pulling fs layer
3531af2bc2a9: Pulling fs layer
b05032fa5b62: Pulling fs layer
b4e72df0ba0e: Pulling fs layer
706754b49547: Download complete
d8377d858729: Download complete
6f7b1a1af1d2: Download complete
b4e72df0ba0e: Download complete
b05032fa5b62: Download complete
3531af2bc2a9: Download complete
3531af2bc2a9: Pull complete
b4e72df0ba0e: Pull complete
6f7b1a1af1d2: Pull complete
b05032fa5b62: Pull complete
Digest: sha256:cdbf8193cee2e31639ea8ea85ffdd8fa5cce98ee9abfde96ea5f329490048831
Status: Downloaded newer image for python:3.10-slim
 ---> cdbf8193cee2
Step 2/5 : WORKDIR /app
 ---> Running in 472e58e70b5b
 ---> Removed intermediate container 472e58e70b5b
 ---> f024f3f6c55b
Step 3/5 : RUN pip install kafka-python
 ---> Running in 760b21982dff
Collecting kafka-python
  Downloading kafka_python-2.3.1-py2.py3-none-any.whl (326 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 326.1/326.1 kB 14.7 MB/s eta 0:00:00
Installing collected packages: kafka-python
Successfully installed kafka-python-2.3.1
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manag
er. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv

[notice] A new release of pip is available: 23.0.1 -> 26.1.1
[notice] To update, run: pip install --upgrade pip
 ---> Removed intermediate container 760b21982dff
 ---> d00551c3631f
Step 4/5 : COPY producer.py .
 ---> f6c59bf81a76
Step 5/5 : CMD ["python", "producer.py"]
 ---> Running in 8f89a2924d95
 ---> Removed intermediate container 8f89a2924d95
 ---> 2fbaa1debf63
Successfully built 2fbaa1debf63
Successfully tagged eurovision-producer:latest
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$
```

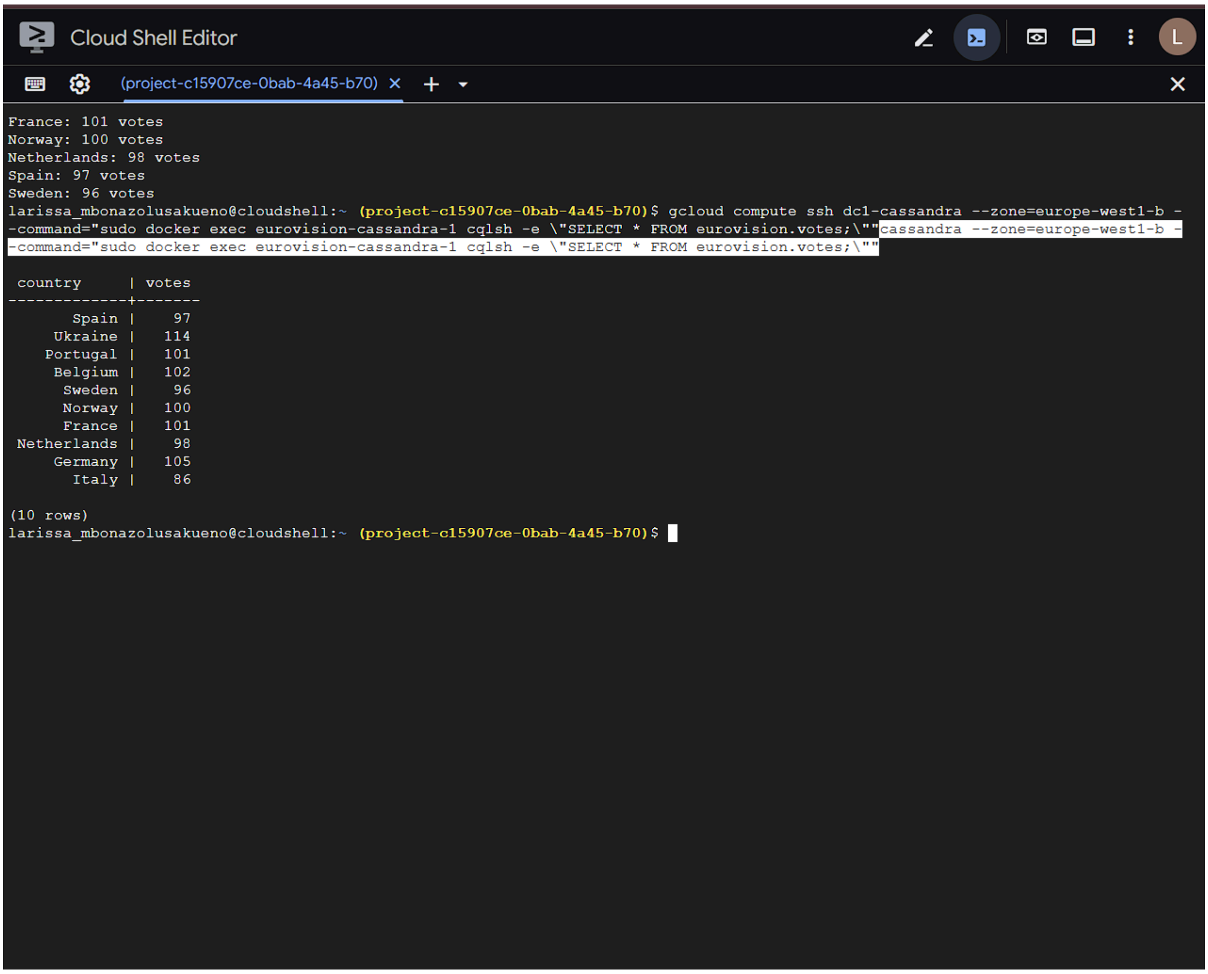
```
Vote 956: Belgium
Vote 957: Spain
Vote 958: Norway
Vote 959: France
Vote 960: Sweden
Vote 961: Netherlands
Vote 962: Netherlands
Vote 963: Ukraine
Vote 964: Portugal
Vote 965: Portugal
Vote 966: Germany
Vote 967: Ukraine
Vote 968: Portugal
Vote 969: Spain
Vote 970: Ukraine
Vote 971: Sweden
Vote 972: Sweden
Vote 973: Portugal
Vote 974: Spain
Vote 975: Germany
Vote 976: Germany
Vote 977: France
Vote 978: France
Vote 979: Belgium
Vote 980: Norway
Vote 981: Germany
Vote 982: Norway
Vote 983: Belgium
Vote 984: Norway
Vote 985: Portugal
Vote 986: Belgium
Vote 987: Norway
Vote 988: Portugal
Vote 989: Italy
Vote 990: Italy
Vote 991: Norway
Vote 992: Ukraine
Vote 993: Italy
Vote 994: Belgium
Vote 995: Portugal
Vote 996: Sweden
Vote 997: Belgium
Vote 998: Ukraine
Vote 999: Belgium
Vote 1000: Belgium
Done! 1000 votes sent.
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$
```

```
3531af2bc2a9: Pull complete
b4e72df0ba0e: Pull complete
6f7b1a1af1d2: Pull complete
b05032fa5b62: Pull complete
Digest: sha256:cdbf8193cee2e31639ea8ea85ffdd8fa5cce98ee9abfde96ea5f329490048831
Status: Downloaded newer image for python:3.10-slim
 ---> cdbf8193cee2
Step 2/5 : WORKDIR /app
 ---> Running in 7f2ccb9ecdcc
 ---> Removed intermediate container 7f2ccb9ecdcc
 ---> 9110441fdff4
Step 3/5 : RUN pip install kafka-python cassandra-driver
 ---> Running in a59e5b2932a7
Collecting kafka-python
  Downloading kafka_python-2.3.1-py2.py3-none-any.whl (326 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 326.1/326.1 kB 15.4 MB/s eta 0:00:00
Collecting cassandra-driver
  Downloading cassandra_driver-3.30.0-cp310-cp310-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (7.8 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 7.8/7.8 MB 19.2 MB/s eta 0:00:00
Collecting geomet>=1.1
  Downloading geomet-1.1.0-py3-none-any.whl (31 kB)
Collecting Deprecated>=1.3.1
  Downloading deprecated-1.3.1-py2.py3-none-any.whl (11 kB)
Collecting wrapt<3,>=1.10
  Downloading wrapt-2.1.2-cp310-cp310-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (113 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 113.6/113.6 kB 11.9 MB/s eta 0:00:00
Collecting click
  Downloading click-8.3.3-py3-none-any.whl (110 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 110.5/110.5 kB 14.0 MB/s eta 0:00:00
Installing collected packages: kafka-python, wrapt, click, geomet, Deprecated, cassandra-driver
Successfully installed Deprecated-1.3.1 cassandra-driver-3.30.0 click-8.3.3 geomet-1.1.0 kafka-python-2.3.1 wrapt-2.1.2
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manag
er. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv

[notice] A new release of pip is available: 23.0.1 -> 26.1.1
[notice] To update, run: pip install --upgrade pip
 ---> Removed intermediate container a59e5b2932a7
 ---> 9017d62a550c
Step 4/5 : COPY spark_job.py .
 ---> 305d28406234
Step 5/5 : CMD ["python", "spark_job.py"]
 ---> Running in f776a98496e4
 ---> Removed intermediate container f776a98496e4
 ---> b7e58ceb251d
Successfully built b7e58ceb251d
Successfully tagged eurovision-spark:latest
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$
```

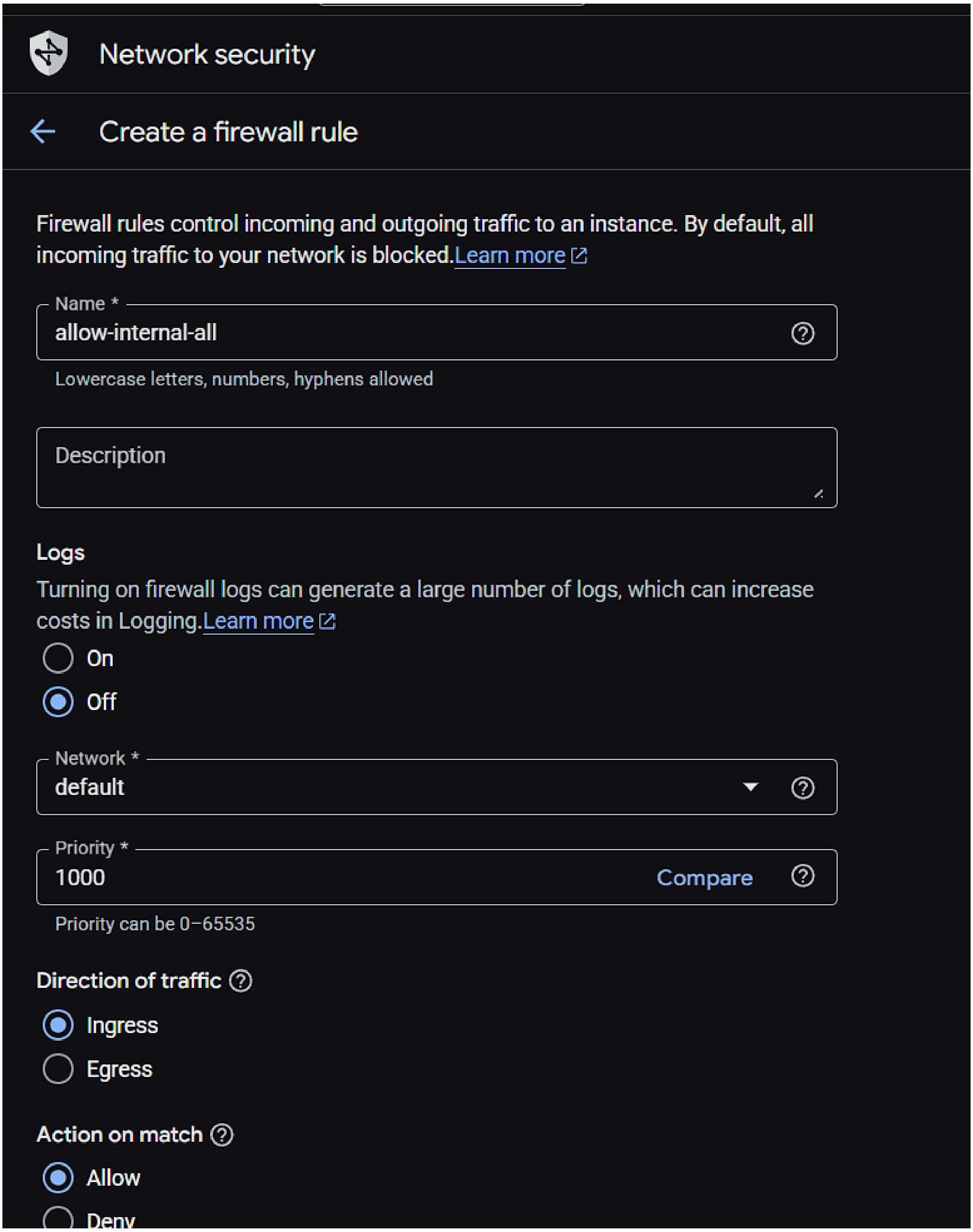
Cloud Shell Editor

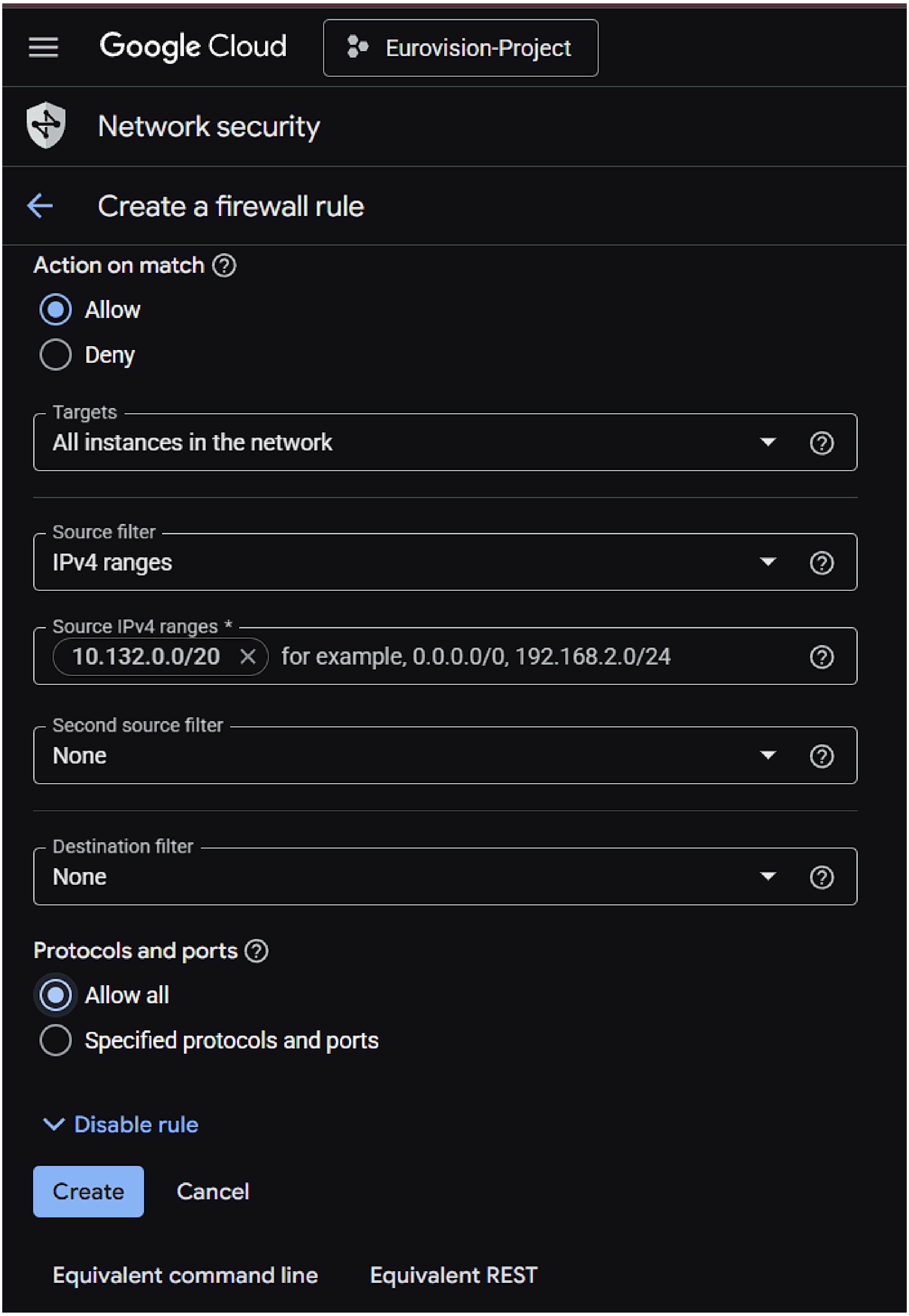
(project-c15907ce-0bab-4a45-b70)

```
park_job.py"    session.execute(\"INSERT INTO votes (country, votes) VALUES (%s, %s)\", (country, cnt))' >> ~/eurovision/spark/s
park_job.py"
                                                                       gcloud compute ssh dc1-spark --zone=europe-west1-b --com
mand="echo 'print(\"Saved to Cassandra!\")' >> ~/eurovision/spark/spark_job.py"compute ssh dc1-spark --zone=europe-west1-b --com
-bash: !\: event not foundto Cassandra!\")' >> ~/eurovision/spark/spark_job.py"
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute ssh dc1-spark --zone=europe-west1-b --com
mand="cd ~/eurovision/spark && sudo docker build -t eurovision-spark ." gcloud compute ssh dc1-spark --zone=europe-west1-b --com
mand="cd ~/eurovision/spark && sudo docker build -t eurovision-spark ."


DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.584kB
Step 1/5 : FROM python:3.10-slim
 ---> cdbf8193cee2
Step 2/5 : WORKDIR /app
 ---> Using cache
 ---> 9110441fdff4
Step 3/5 : RUN pip install kafka-python cassandra-driver
 ---> Using cache
 ---> 9017d62a550c
Step 4/5 : COPY spark_job.py .
 ---> 21b22f59ab39
Step 5/5 : CMD ["python", "spark_job.py"]
 ---> Running in e6ee5190f03f
 ---> Removed intermediate container e6ee5190f03f
 ---> e398b9532670
Successfully built e398b9532670
Successfully tagged eurovision-spark:latest                            gcloud compute ssh dc1-spark --zone=europe-west1-b --com
mand="sudo docker run --rm --network host eurovision-spark"b-4a45-b70)$ gcloud compute ssh dc1-spark --zone=europe-west1-b --com
mand="sudo docker run --rm --network host eurovision-spark"
Results:
Ukraine: 114 votes
Germany: 105 votes
Belgium: 102 votes
Portugal: 101 votes
France: 101 votes
Norway: 100 votes
Netherlands: 98 votes
Spain: 97 votes
Sweden: 96 votes
Italy: 86 votes
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$
```

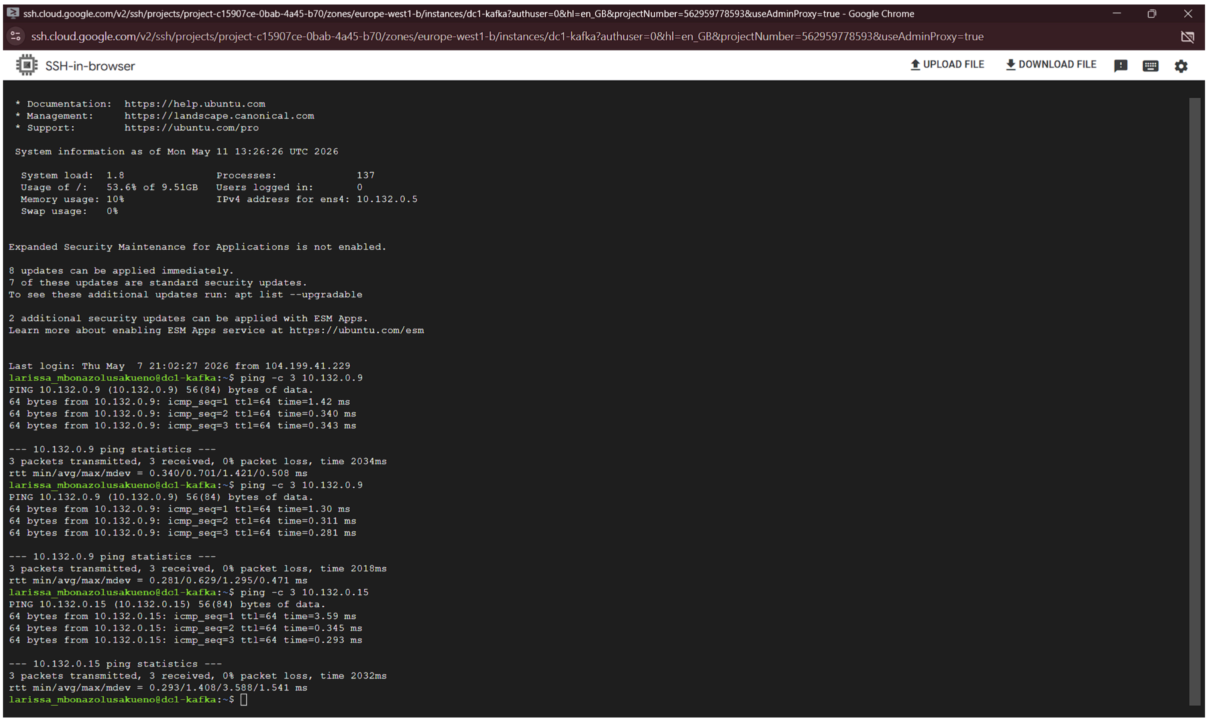


firewall opzetten


Network security
Create a firewall rule
Firewall rules control incoming and outgoing traffic to an instance. By default, all incoming traffic to your network is blocked. Learn more
Name *
allow-internal-all
Lowercase letters, numbers, hyphens allowed
Description
Logs
Turning on firewall logs can generate a large number of logs, which can increase costs in Logging. Learn more
On
Off
Network *
default
Priority *
1000
Compare
Priority can be 0–65535
Direction of traffic
Ingress
Egress
Action on match
Allow
Deny


Google Cloud
Eurovision-Project
Network security
Create a firewall rule
Action on match
Allow
Deny
Targets
All instances in the network
Source filter
IPv4 ranges
Source IPv4 ranges *
10.132.0.0/20
for example, 0.0.0.0/0, 192.168.2.0/24
Second source filter
None
Destination filter
None
Protocols and ports
Allow all
Specified protocols and ports
Disable rule
Create
Cancel
Equivalent command line
Equivalent REST

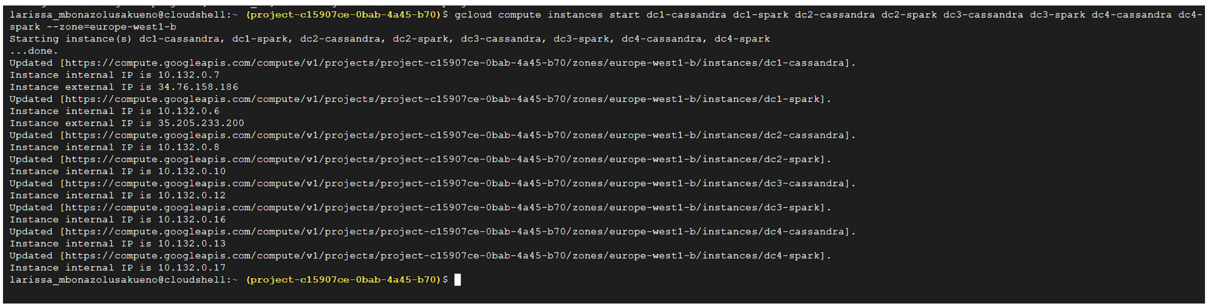
connectiviteit tussen de vm's ()

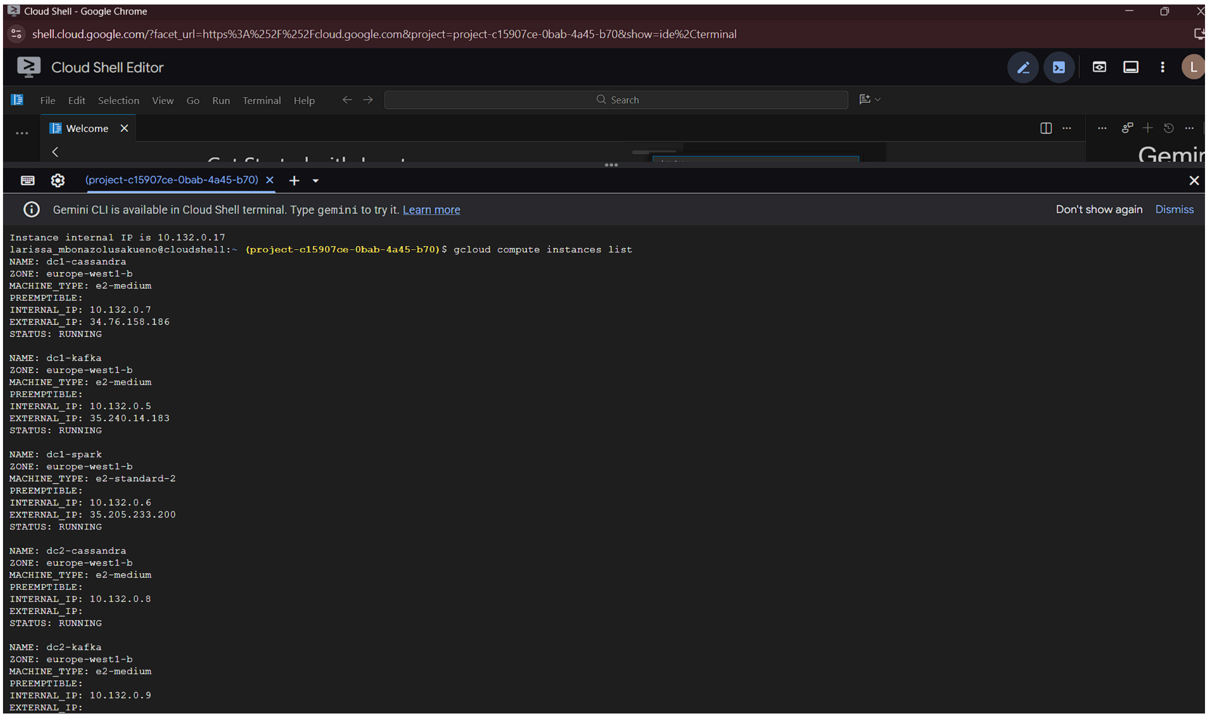


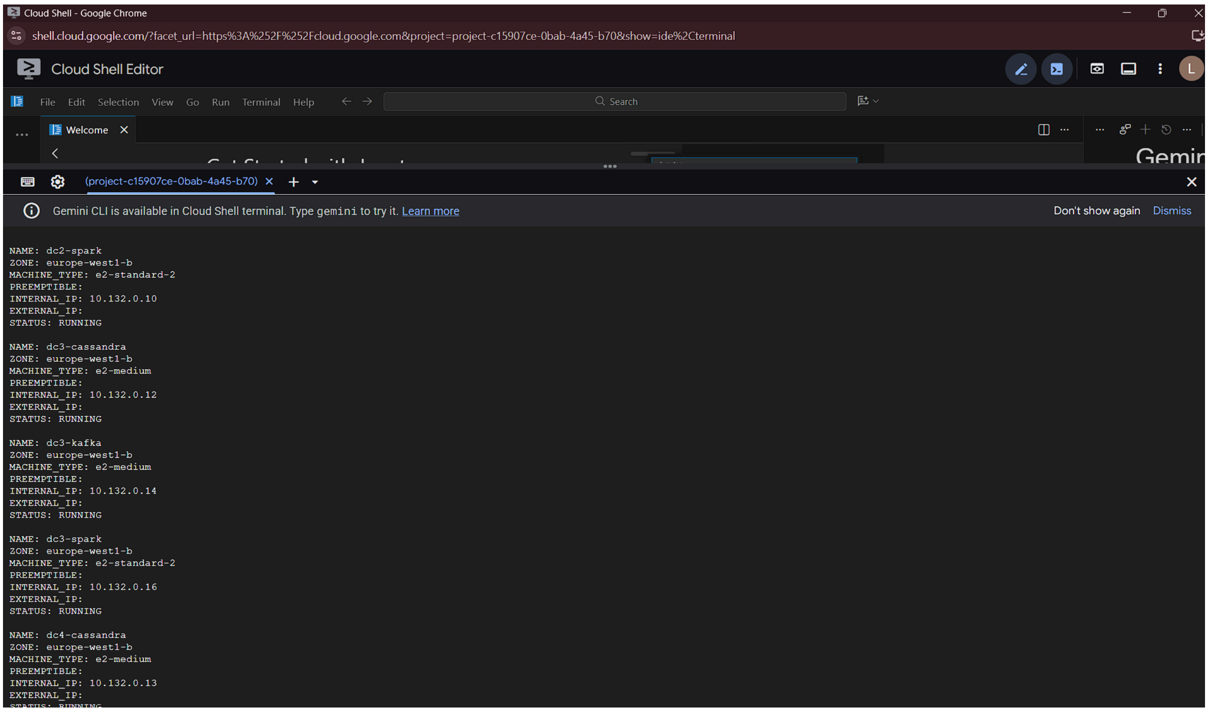
Eerst alle VM's opstarten:

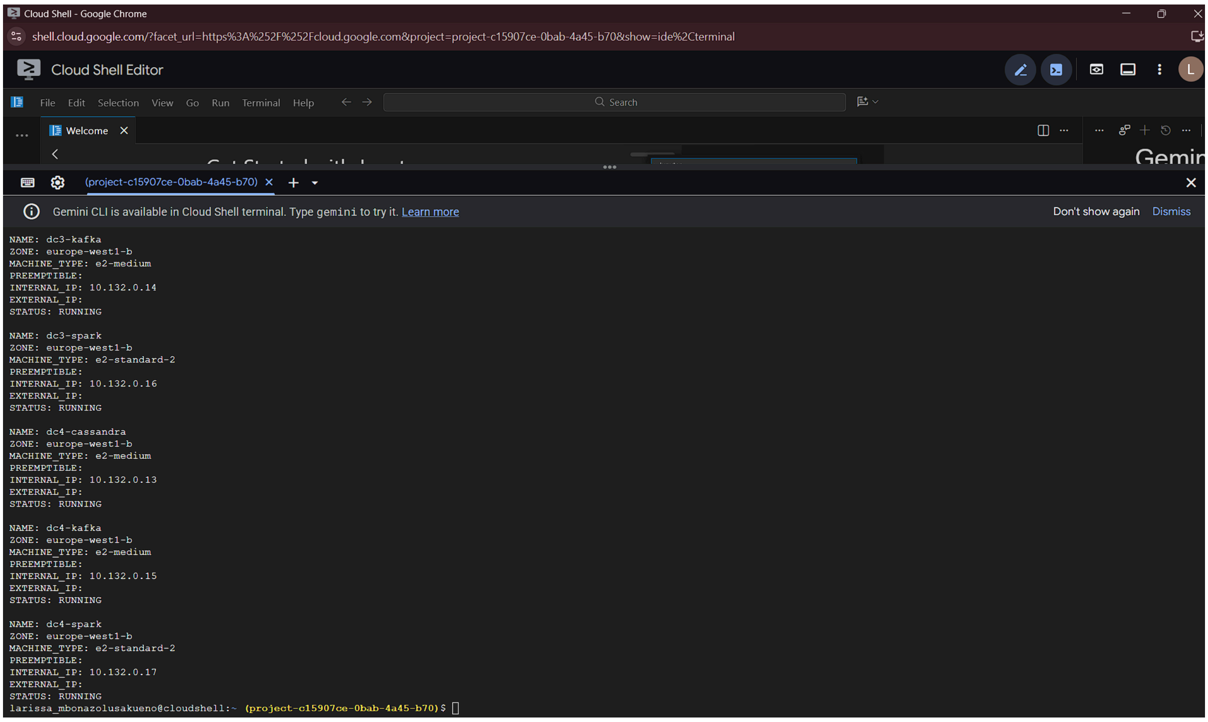
bash

gcloud compute instances start dc1-cassandra dc1-spark dc2-cassandra dc2-spark dc3-c

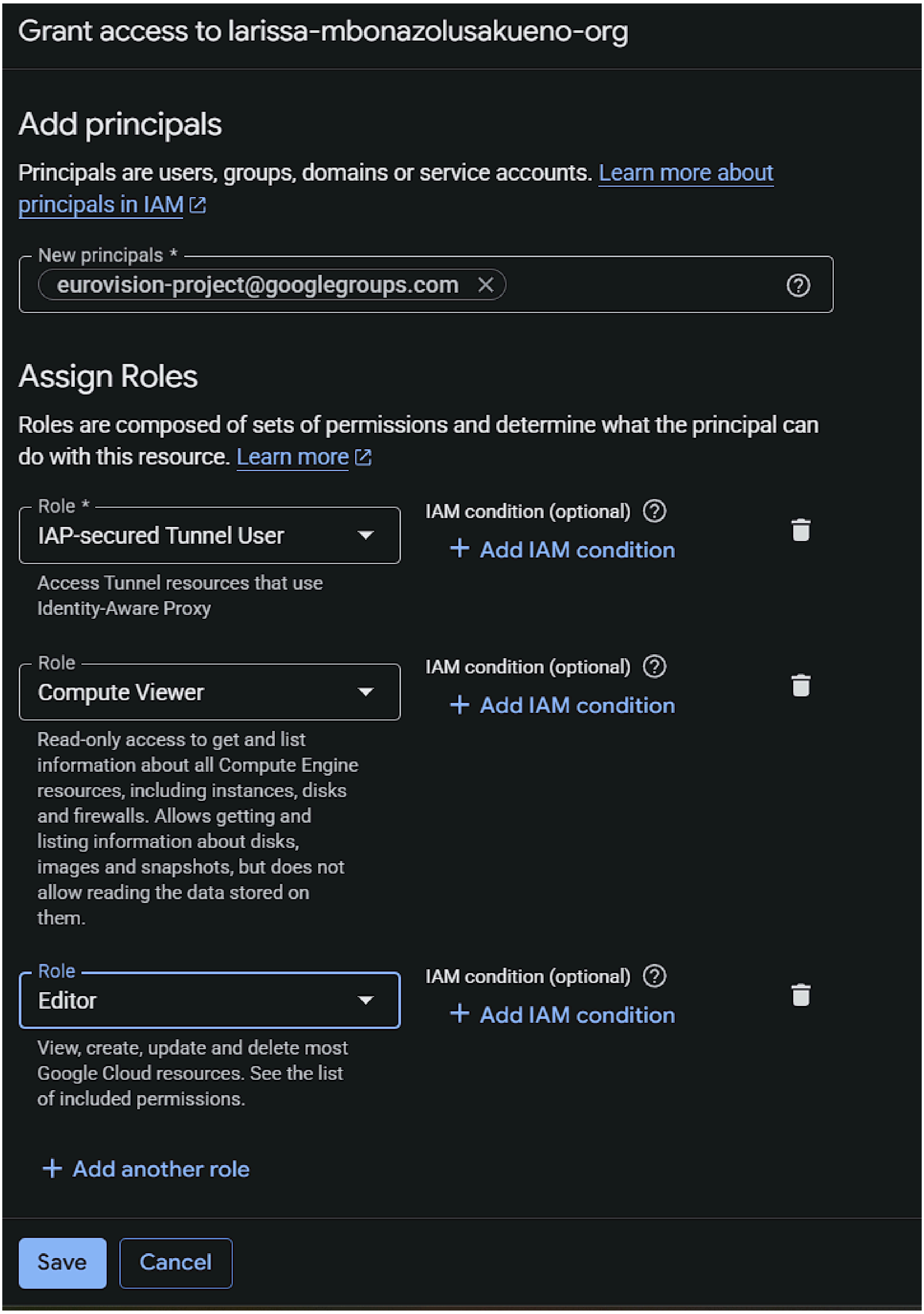


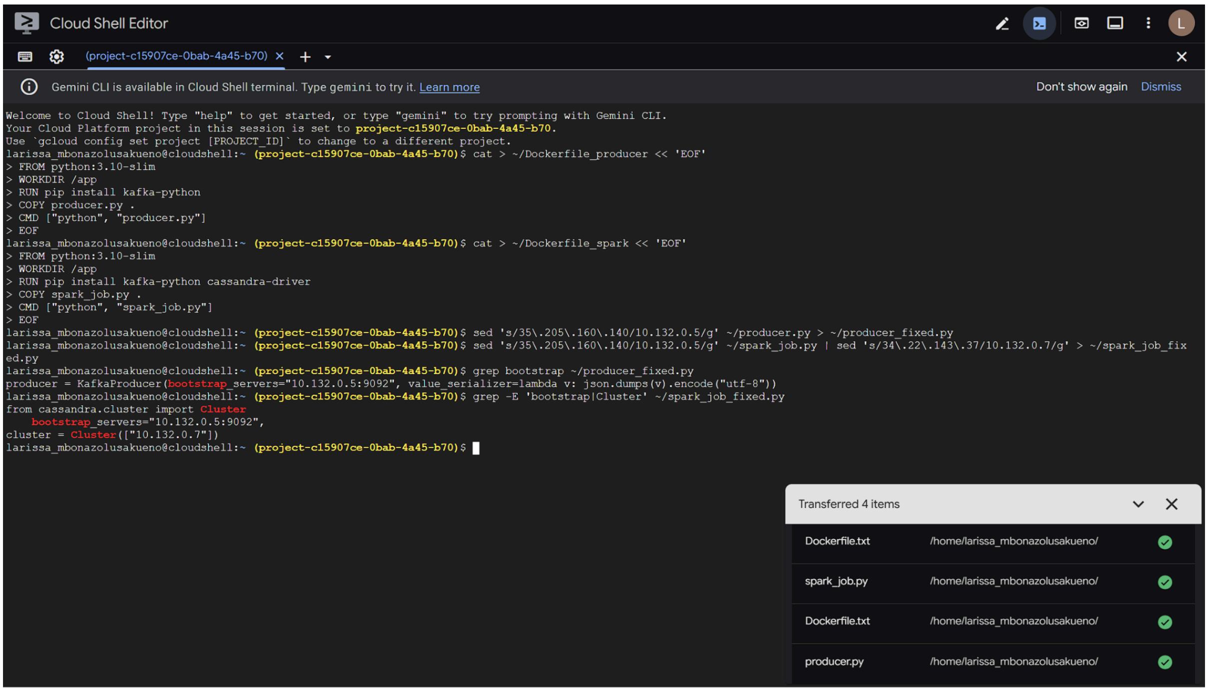

Cloud Shell - Google Chrome
shell.cloud.google.com/?facet_url=https%3A%252F%252Fcloud.google.com&project=project-c15907ce-0bab-4a45-b70&show=ide%2Cterminal
Cloud Shell Editor
File Edit Selection View Go Run Terminal Help
Search
Welcome
(project-c15907ce-0bab-4a45-b70)
Gemini CLI is available in Cloud Shell terminal. Type gemini to try it. Learn more
Don't show again Dismiss
Instance internal IP is 10.132.0.17
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ gcloud compute instances list
NAME: dc1-cassandra
ZONE: europe-west1-b
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.7
EXTERNAL_IP: 34.76.158.186
STATUS: RUNNING
NAME: dc1-kafka
ZONE: europe-west1-b
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.5
EXTERNAL_IP: 35.240.14.183
STATUS: RUNNING
NAME: dc1-spark
ZONE: europe-west1-b
MACHINE_TYPE: e2-standard-2
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.6
EXTERNAL_IP: 35.205.233.200
STATUS: RUNNING
NAME: dc2-cassandra
ZONE: europe-west1-b
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.8
EXTERNAL_IP:
STATUS: RUNNING
NAME: dc2-kafka
ZONE: europe-west1-b
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.9
EXTERNAL_IP:


Cloud Shell - Google Chrome
shell.cloud.google.com/?facet_url=https%3A%252F%252Fcloud.google.com&project=project-c15907ce-0bab-4a45-b70&show=ide%2Cterminal
Cloud Shell Editor
File Edit Selection View Go Run Terminal Help
Search
Welcome
(project-c15907ce-0bab-4a45-b70)
Gemini CLI is available in Cloud Shell terminal. Type gemini to try it. Learn more
Don't show again Dismiss
NAME: dc2-spark
ZONE: europe-west1-b
MACHINE_TYPE: e2-standard-2
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.10
EXTERNAL_IP:
STATUS: RUNNING
NAME: dc3-cassandra
ZONE: europe-west1-b
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.12
EXTERNAL_IP:
STATUS: RUNNING
NAME: dc3-kafka
ZONE: europe-west1-b
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.14
EXTERNAL_IP:
STATUS: RUNNING
NAME: dc3-spark
ZONE: europe-west1-b
MACHINE_TYPE: e2-standard-2
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.16
EXTERNAL_IP:
STATUS: RUNNING
NAME: dc4-cassandra
ZONE: europe-west1-b
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.13
EXTERNAL_IP:



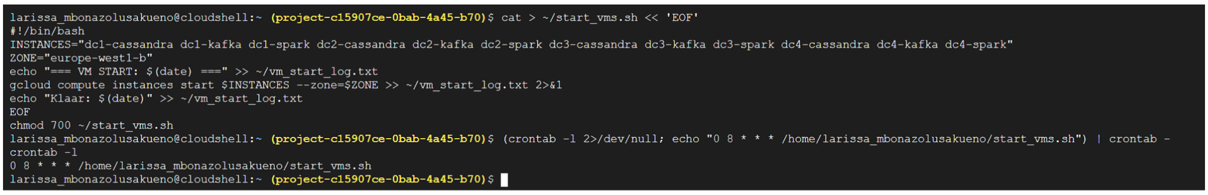
Cloud Shell - Google Chrome
shell.cloud.google.com/?facet_url=https%3A%252F%252Fcloud.google.com&project=project-c15907ce-0bab-4a45-b70&show=ide%2Cterminal
Cloud Shell Editor
File Edit Selection View Go Run Terminal Help
Search
Welcome
(project-c15907ce-0bab-4a45-b70)
Gemini CLI is available in Cloud Shell terminal. Type gemini to try it. Learn more
Don't show again Dismiss
NAME: dc3-kafka
ZONE: europe-west1-b
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.14
EXTERNAL_IP:
STATUS: RUNNING
NAME: dc3-spark
ZONE: europe-west1-b
MACHINE_TYPE: e2-standard-2
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.16
EXTERNAL_IP:
STATUS: RUNNING
NAME: dc4-cassandra
ZONE: europe-west1-b
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.13
EXTERNAL_IP:
STATUS: RUNNING
NAME: dc4-kafka
ZONE: europe-west1-b
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.15
EXTERNAL_IP:
STATUS: RUNNING
NAME: dc4-spark
ZONE: europe-west1-b
MACHINE_TYPE: e2-standard-2
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.17
EXTERNAL_IP:
STATUS: RUNNING
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$


Grant access to larissa-mbonazolusakueno-org
Add principals
Principals are users, groups, domains or service accounts. Learn more about principals in IAM
New principals *
eurovision-project@googlegroups.com
Assign Roles
Roles are composed of sets of permissions and determine what the principal can do with this resource. Learn more
Role *
IAP-secured Tunnel User
Access Tunnel resources that use Identity-Aware Proxy
IAM condition (optional)
+ Add IAM condition
Role
Compute Viewer
Read-only access to get and list information about all Compute Engine resources, including instances, disks and firewalls. Allows getting and listing information about disks, images and snapshots, but does not allow reading the data stored on them.
IAM condition (optional)
+ Add IAM condition
Role
Editor
View, create, update and delete most Google Cloud resources. See the list of included permissions.
IAM condition (optional)
+ Add IAM condition
+ Add another role
Save
Cancel




```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
Your Cloud Platform project in this session is set to project-c15907ce-0bab-4a45-b70.
Use `gcloud config set project [PROJECT_ID]` to change to a different project.
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ cat > ~/Dockerfile_producer << 'EOF'
> FROM python:3.10-slim
> WORKDIR /app
> RUN pip install kafka-python
> COPY producer.py .
> CMD ["python", "producer.py"]
> EOF
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ cat > ~/Dockerfile_spark << 'EOF'
> FROM python:3.10-slim
> WORKDIR /app
> RUN pip install kafka-python cassandra-driver
> COPY spark_job.py .
> CMD ["python", "spark_job.py"]
> EOF
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ sed 's/35\.205\.160\.140/10.132.0.5/g' ~/producer.py > ~/producer_fixed.py
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ sed 's/35\.205\.160\.140/10.132.0.5/g' ~/spark_job.py | sed 's/34\.22\.143\.37/10.132.0.7/g' > ~/spark_job_fix
ed.py
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ grep bootstrap ~/producer_fixed.py
producer = KafkaProducer(bootstrap_servers="10.132.0.5:9092", value_serializer=lambda v: json.dumps(v).encode("utf-8"))
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ grep -E 'bootstrap|Cluster' ~/spark_job_fixed.py
from cassandra.cluster import Cluster
    bootstrap_servers="10.132.0.5:9092",
cluster = Cluster(["10.132.0.7"])
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$
```

automatisch opstarten vd vm's



```
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ cat > ~/start_vms.sh << 'EOF'
#!/bin/bash
INSTANCES="dc1-cassandra dc1-kafka dc1-spark dc2-cassandra dc2-kafka dc2-spark dc3-cassandra dc3-kafka dc3-spark dc4-cassandra dc4-kafka dc4-spark"
ZONE="europe-west1-b"
echo "=== VM START: $(date) ===" >> ~/vm_start_log.txt
gcloud compute instances start $INSTANCES --zone=$ZONE >> ~/vm_start_log.txt 2>&1
echo "Klaar: $(date)" >> ~/vm_start_log.txt
EOF
chmod 700 ~/start_vms.sh
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$ (crontab -l 2>/dev/null; echo "0 8 * * * /home/larissa_mbonazolusakueno/start_vms.sh") | crontab -
crontab -l
0 8 * * * /home/larissa_mbonazolusakueno/start_vms.sh
larissa_mbonazolusakueno@cloudshell:~ (project-c15907ce-0bab-4a45-b70)$
```

Demo stap 1 — Toon alle 12 VM's:

bash

gcloud compute instances list

Demo stap 2 — Reset Cassandra:

bash

gcloud compute ssh dc1-cassandra --zone=europe-west1-b \

--command="sudo docker exec eurovision-cassandra cqlsh -e \"TRUNCATE eurovision.votes;\""

Demo stap 3 — Alle 4 producers:

bash

gcloud compute ssh dc1-kafka --zone=europe-west1-b \

--command="sudo docker run --rm --network host eurovision-producer"

bash

gcloud compute ssh dc2-kafka --zone=europe-west1-b --tunnel-through-iap \

--command="sudo docker run --rm --network host eurovision-producer"

bash

gcloud compute ssh dc3-kafka --zone=europe-west1-b --tunnel-through-iap \

--command="sudo docker run --rm --network host eurovision-producer"

bash

gcloud compute ssh dc4-kafka --zone=europe-west1-b --tunnel-through-iap \

--command="sudo docker run --rm --network host eurovision-producer"

Demo stap 4 — Alle 4 Spark jobs:

bash

gcloud compute ssh dc1-spark --zone=europe-west1-b --tunnel-through-iap \

--command="sudo docker run --rm --network host eurovision-spark"

bash

gcloud compute ssh dc2-spark --zone=europe-west1-b --tunnel-through-iap \

--command="sudo docker run --rm --network host eurovision-spark"

bash

gcloud compute ssh dc3-spark --zone=europe-west1-b --tunnel-through-iap \

--command="sudo docker run --rm --network host eurovision-spark"

bash

gcloud compute ssh dc4-spark --zone=europe-west1-b --tunnel-through-iap \

--command="sudo docker run --rm --network host eurovision-spark"

Demo stap 5 — Eindresultaat:

bash

gcloud compute ssh dc1-cassandra --zone=europe-west1-b \

--command="sudo docker exec eurovision-cassandra cqlsh -e \"SELECT count(*) as totaal_stemmen FROM eurovision.votes;\""

Demo stap 6 — Toon automatisch opstarten (cron):

bash

crontab -l

cat ~/start_vms.sh

Voer ze één voor één uit tijdens de opname en stuur de eindoutput van stap 5!

Literatuurlijst

Azure. (z.d.). *Wat is een virtuele machine (VM)?* Microsoft Azure. Geraadpleegd op 15 maart 2026, van https://azure.microsoft.com/nl-nl/resources/cloud-computing-dictionary/what-is-a-virtual-machine/

Combell. (z.d.). *Wat is Docker?* Combell Blog. Geraadpleegd op 15 maart 2026, van https://www.combell.com/nl/blog/wat-is-docker/

Command Linux. (z.d.). *Linux Adoption in Fortune 500 Companies*. Geraadpleegd op 15 maart 2026, van https://commandlinux.com/statistics/linux-adoption-in-fortune-500-companies/

Data Science Partners. (z.d.). *Python Pandas: De ultieme gids voor data-analyse*. Geraadpleegd op 15 maart 2026, van https://datasciencepartners.nl/python-pandas/

Datapartner365. (z.d.). *Wat is PySpark?* Geraadpleegd op 15 maart 2026, van https://www.datapartner365.nl/blog/wat-is-pyspark.html

W3Schools. (z.d.). *Python Virtualenv*. Geraadpleegd op 15 maart 2026, van https://www.w3schools.com/python/python_virtualenv.asp

Wikipedia-bijdragers. (z.d.). *Python (programmeertaal)*. Wikipedia, de vrije encyclopedie. Geraadpleegd op 15 maart 2026, van https://nl.wikipedia.org/wiki/Python_(programmeertaal

Docker. (z.d.). *Install Docker Engine on Ubuntu.* Geraadpleegd op 15 maart 2026, van https://docs.docker.com/engine/install/ubuntu/#prerequisites

Visual Studio Code. (z.d.). *Visual Studio Code on Linux.* Geraadpleegd op 23 maart 2026, van

https://code.visualstudio.com/docs/setup/linux#_install-vs-code-on-linux

https://www.atlassian.com/nl/work-management/project-management/proof-of-concept